Computational Neuropathology: Clinical and Research Applications of Computer Vision Models

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UHN Laboratory Medicine Program
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Disclosures

- I have no relevant financial relationships to disclose

Learning Objevtives

1. Discuss concepts of how AI can be used to automate classification and quantitative tasks in neuropathology
2. Discuss how quantitative tissue feature signatures can be applied to predict outcomes in molecular subtypes of brain tumors
3. Describe how quantitative histology can aid in the study of intra-tumoral heterogeneity
4. List limitations and caveats of computational techniques in histological analysis

Outline & Disclaimer

- Semantic Segmentation:
    - Supervised Learning
        - H&E-based tumor detection and classification

- Instance Segmentation:
    - IHC-based : Ki-67 Quantification

- Multi-slide Image registration
    - Merging H&E and IHC-level information

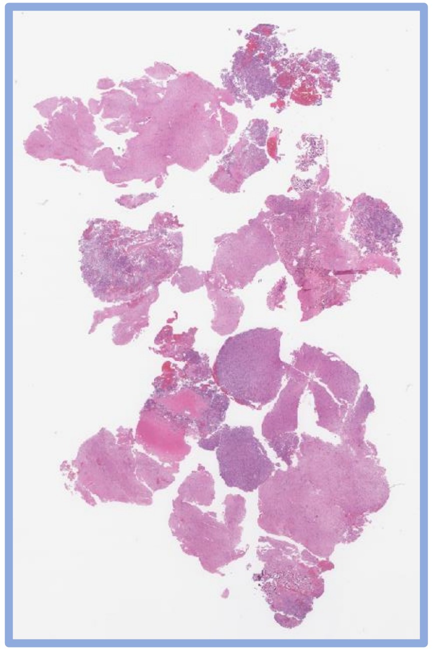
- Additional applications:
    - Unsupervised Learning: Heterogeneity assessment
    - Custom model development & clinical correlation studies
    - PHARAOH (pathologyreports.ai)
    - Integrating and refining molecular subtypes of gliomas

Note of Caution

- Not exhaustive survey of field
- Engineering NOT a knowledge-based field
- Focus on concepts and tools used / developed first-hand and not necessarily on ”Best” point solutions for each problem
- Results vary based on study design

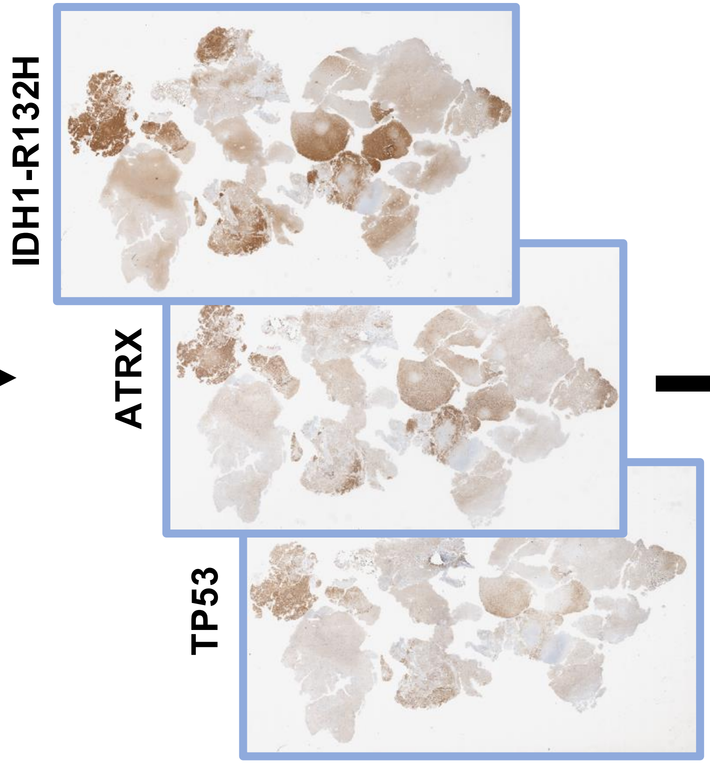
Modern molecular neuropathology workflow

Step 1
Histomorphologic Classification

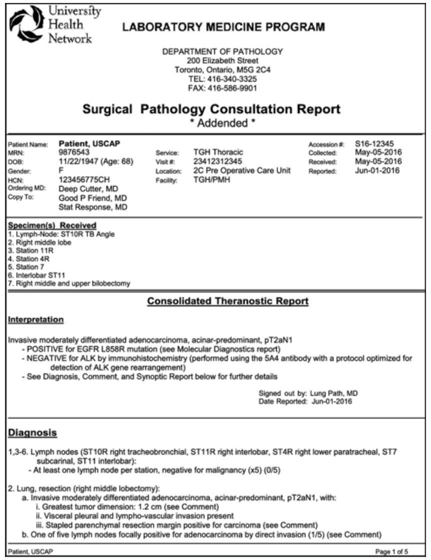


H&E slide

Step 2
Biomarker Interpretation and Integration



Step 3
Integrated Diagnosis



University Health Network
LABORATORY MEDICINE PROGRAM

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200 Elizabeth Street
Toronto, Ontario, M5G 2C4
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Surgical Pathology Consultation Report
* Addended *

Patient Name:	Patient, USCAP			Accession #:	S16-12345
MRN:	9876543	Service:	TGH Thoracic	Collected:	May-05-2016
DOB:	11/22/1947 (Age: 68)	Visit #:	23412312345	Received:	May-05-2016
Gender:	F	Location:	2C Pre Operative Care Unit	Reported:	Jun-01-2016
HCN:	123456775CH	Facility:	TGH/PMH		
Ordering MD:	Deep Cutter, MD				
Copy To:	Good P Friend, MD Stat Response, MD				

Specimen(s) Received
1. Lymph-Node: ST10R TB Angle
2. Right middle lobe
3. Station 11R
4. Station 4R
5. Station 7
6. Interlobar ST11
7. Right middle and upper bilobectomy

Consolidated Theranostic Report

Interpretation

Invasive moderately differentiated adenocarcinoma, acinar-predominant, pT2aN1
- POSITIVE for EGFR L858R mutation (see Molecular Diagnostics report)
- NEGATIVE for ALK by immunohistochemistry (performed using the 5A4 antibody with a protocol optimized for detection of ALK gene rearrangement)
- See Diagnosis, Comment, and Synoptic Report below for further details

Signed out by: Lung Path, MD
Date Reported: Jun-01-2016

Diagnosis

1,3-6. Lymph nodes (ST10R right tracheobronchial, ST11R right interlobar, ST4R right lower paratracheal, ST7 subcarinal, ST11 interlobar):
- At least one lymph node per station, negative for malignancy (x5) (0/5)

2. Lung, resection (right middle lobectomy):
a. Invasive moderately differentiated adenocarcinoma, acinar-predominant, pT2aN1, with:
i. Greatest tumor dimension: 1.2 cm (see Comment)
ii. Visceral pleural and lympho-vascular invasion present
iii. Stapled parenchymal resection margin positive for carcinoma (see Comment)
b. One of five lymph nodes focally positive for adenocarcinoma by direct invasion (1/5) (see Comment)

Patient, USCAP Page 1 of 5

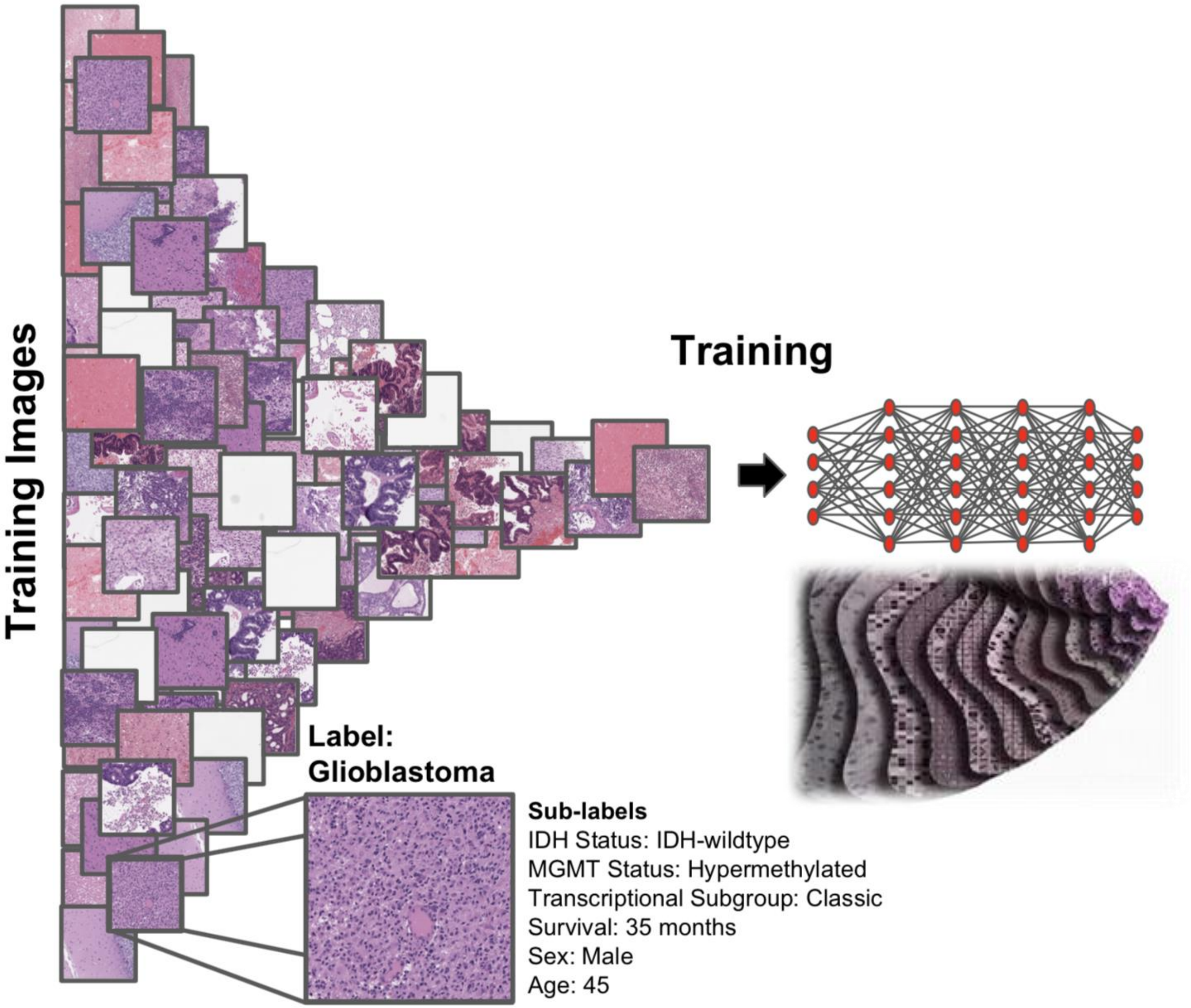
Advanced molecular studies

Advantages:
- Excellent turnaround
- Low tissue requirements
- Cost effective

Disadvantages:
- Many manual steps
- Qualitative

Step 1: Semantic segmentation

Neural Networks - Computer vision tools for automated pattern recognition



Xie & Faust et al, *Crit Rev Lab Sci*, 2019

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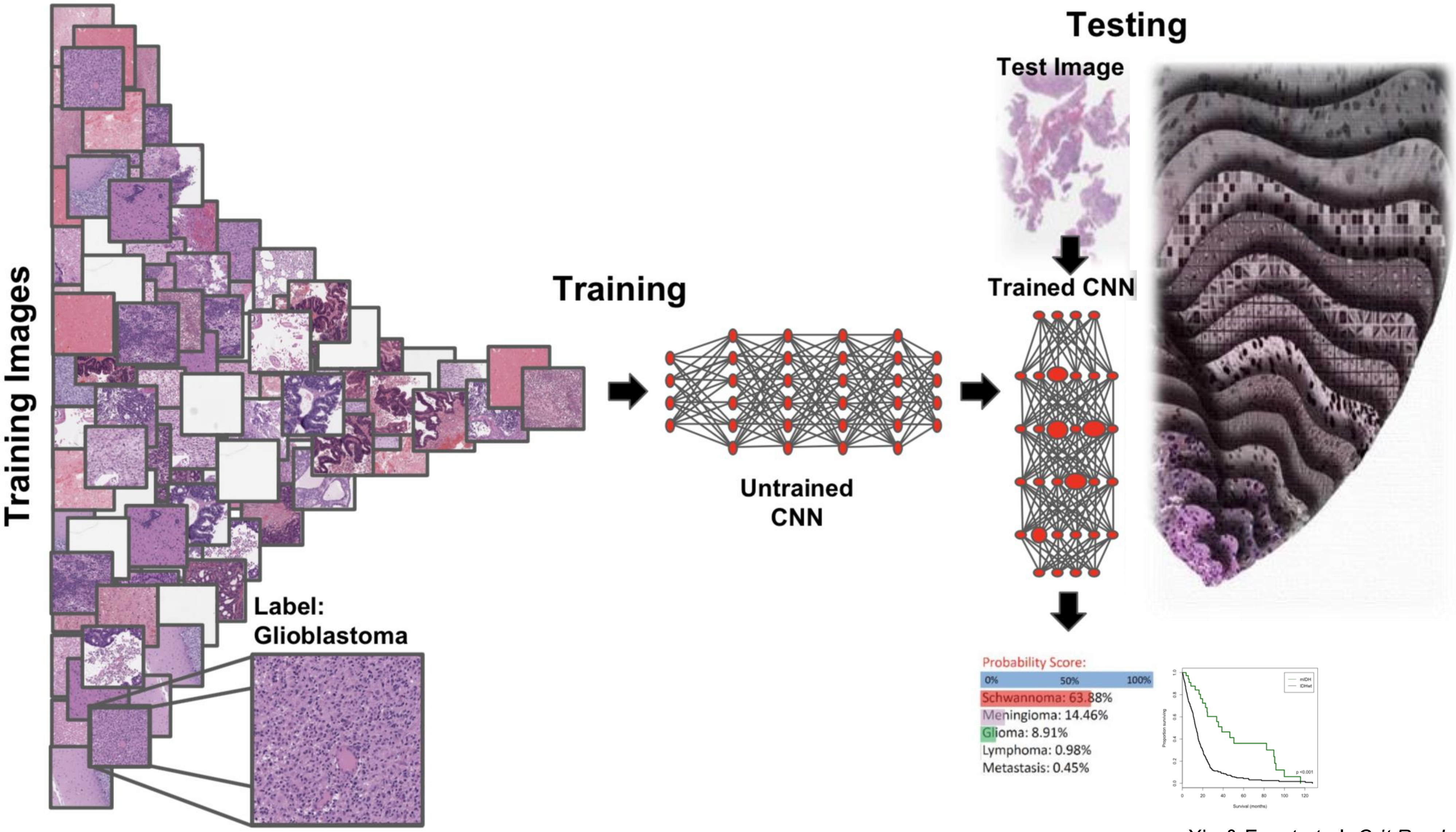
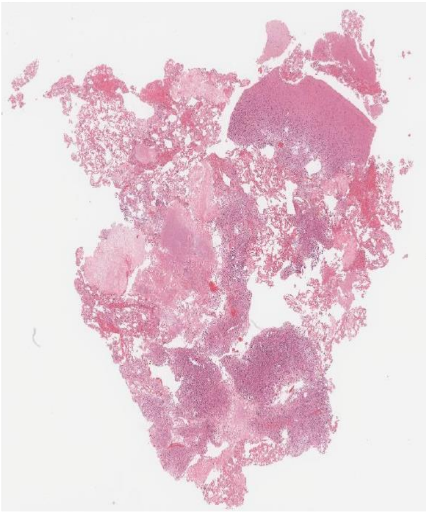
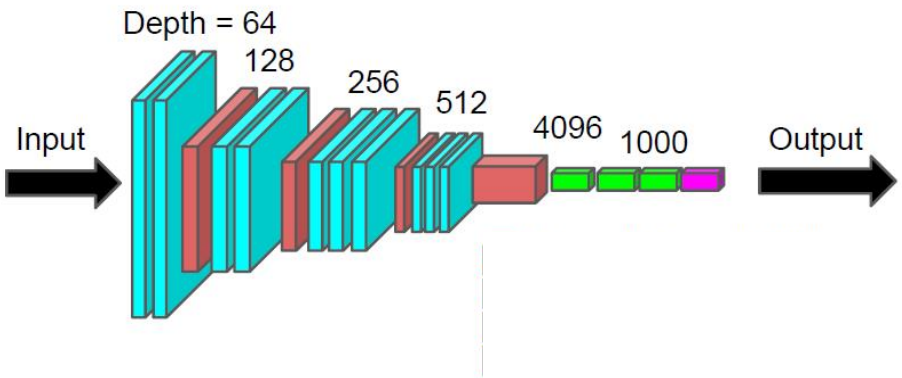
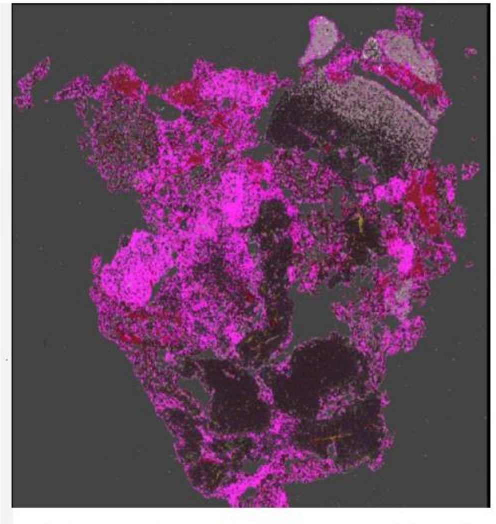


Xie & Faust et al, *Crit Rev Lab Sci*, 2019

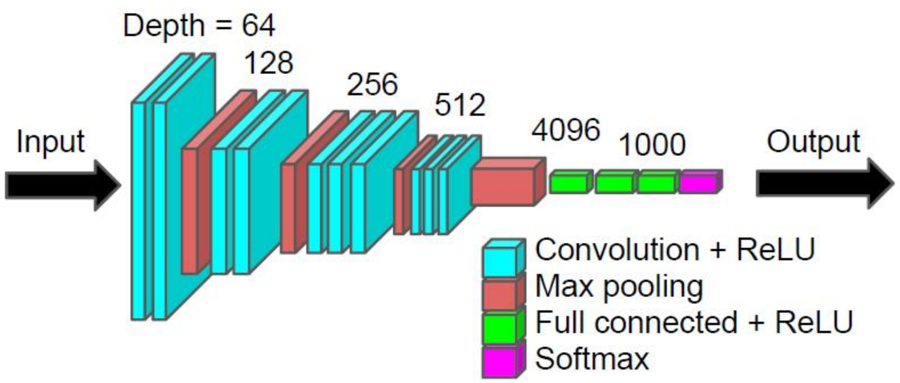
Quantitative semantic segmentation of cellular patterns

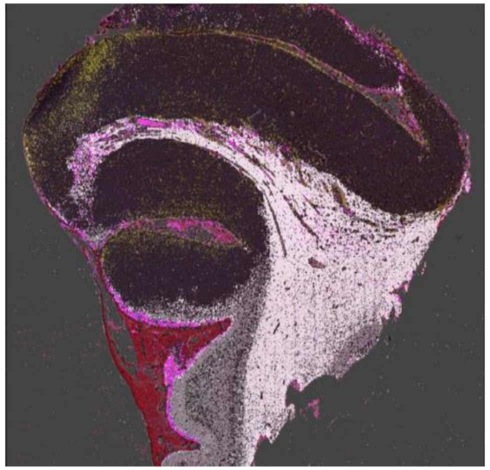






Necrosis	53.54%
Lesional Tissue	28.43%
Gray Matter	7.53%
Blood	6.44%
Surgical Material	3.87%





Lesional Tissue	55.06%
Gray Matter	24.38%
White Matter	24.38%
Blood	5.70%
Necrosis	3.34%

Deep learning-based classification of major brain tumors types

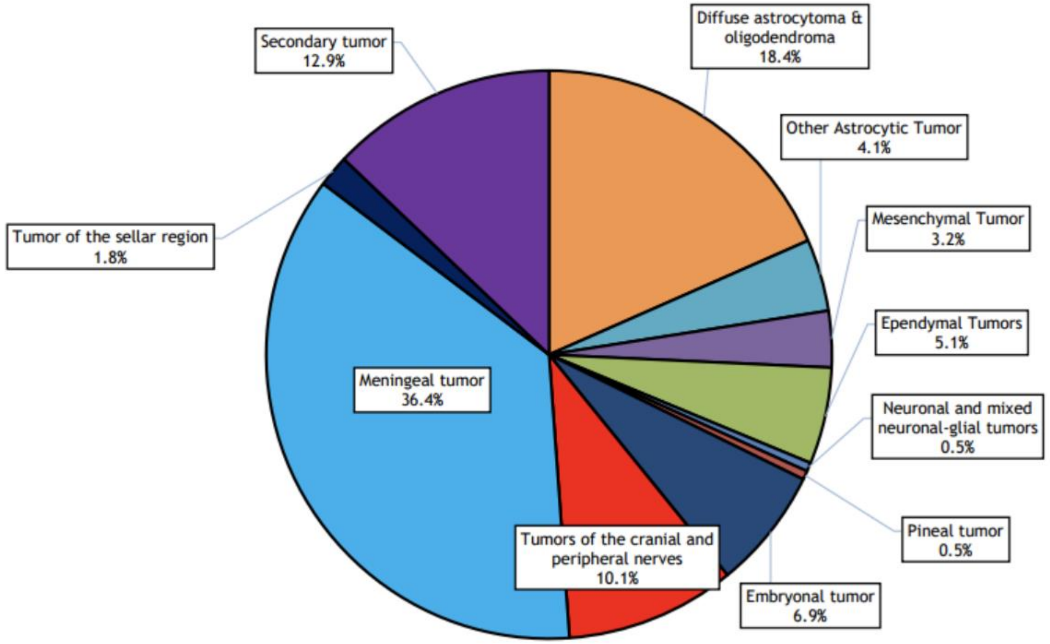
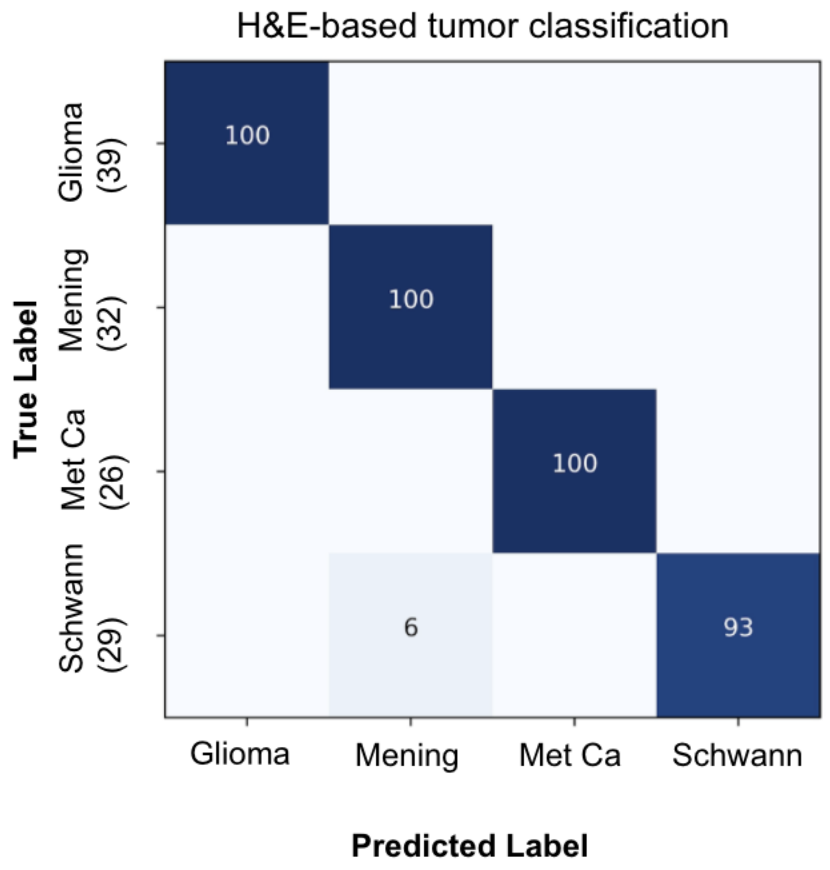


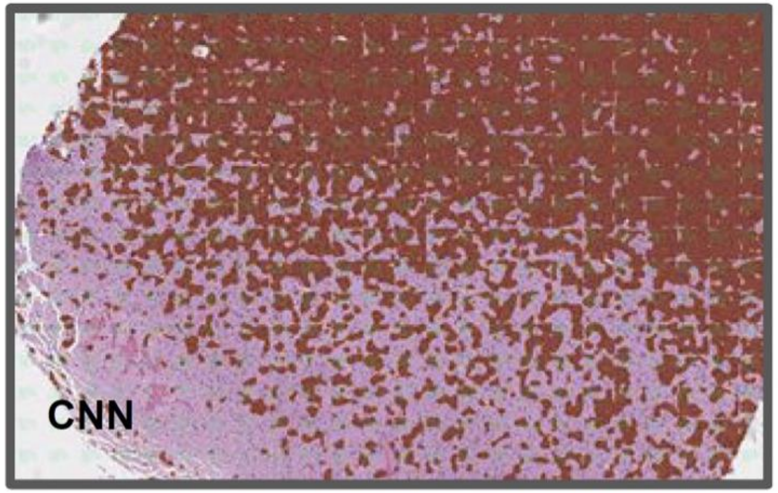
Figure 1: Distribution of major histological types of brain tumour.

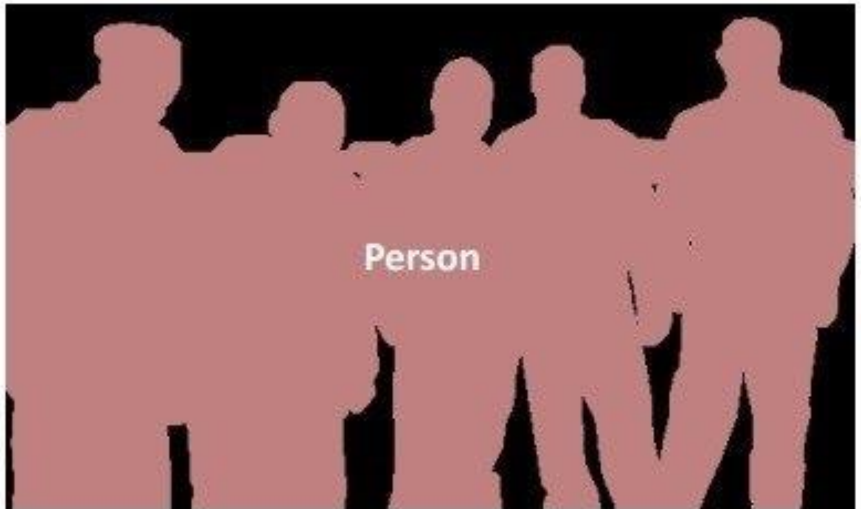
Dzali et al, Mal Jour Pub Health Med 2017

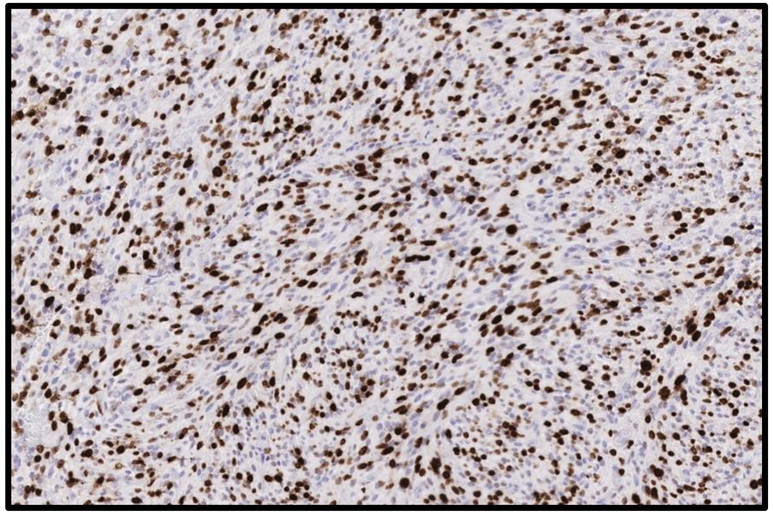


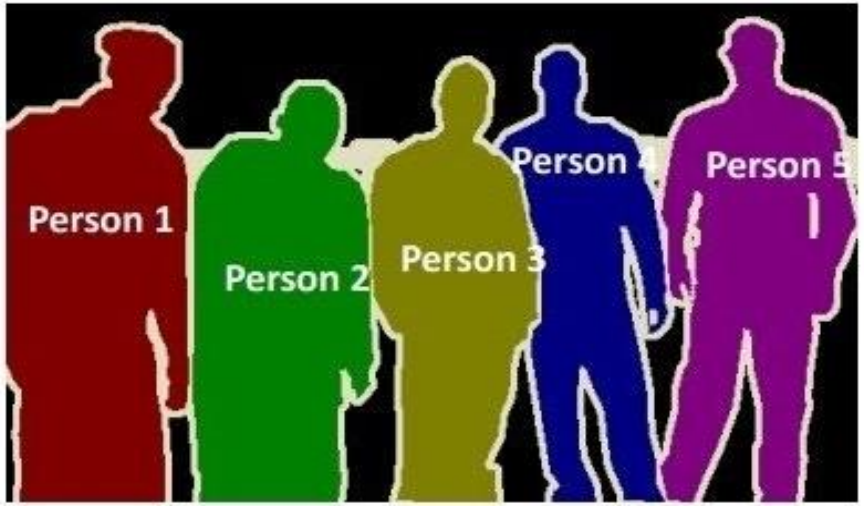
Step 2: Biomarker quantification (Instance segmentation)

Mask R-CNN: Instance segmentation = Semantic segmentation + object detection



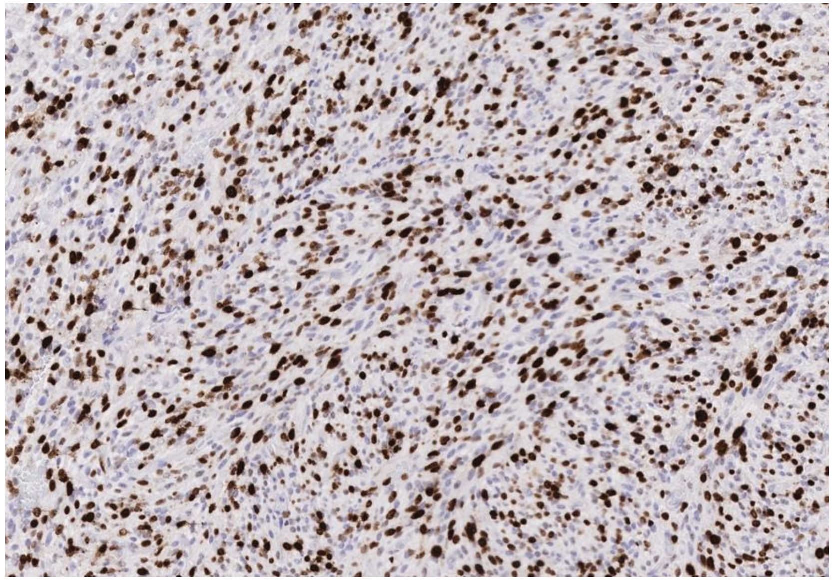


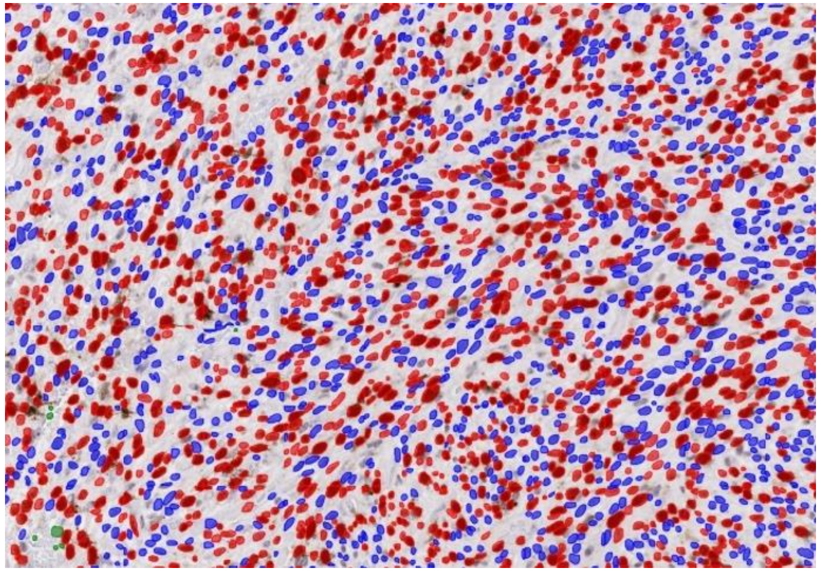


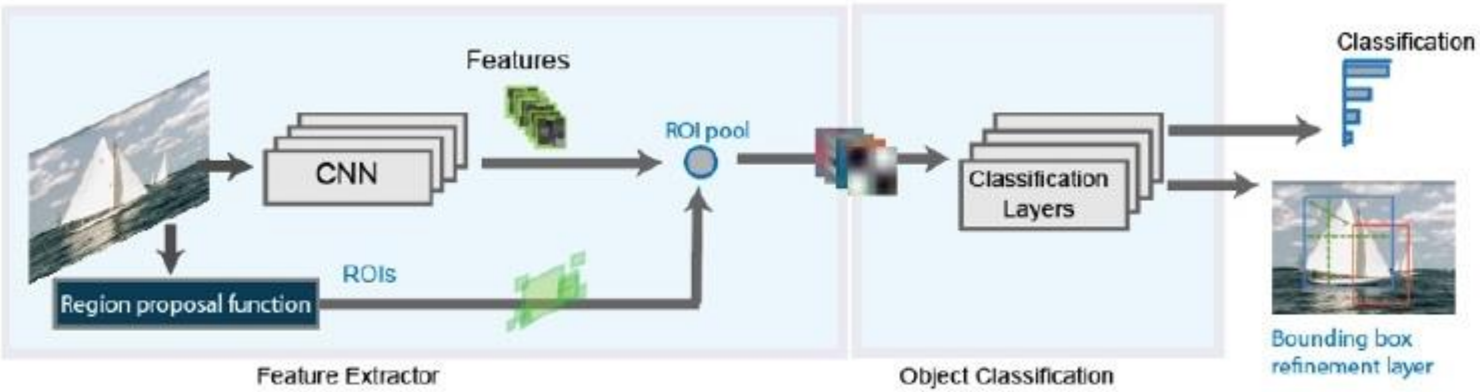


https://viso.ai/deep-learning/mask-r-cnn/

Nuclear segmentation and quantification with Mask R-CNN

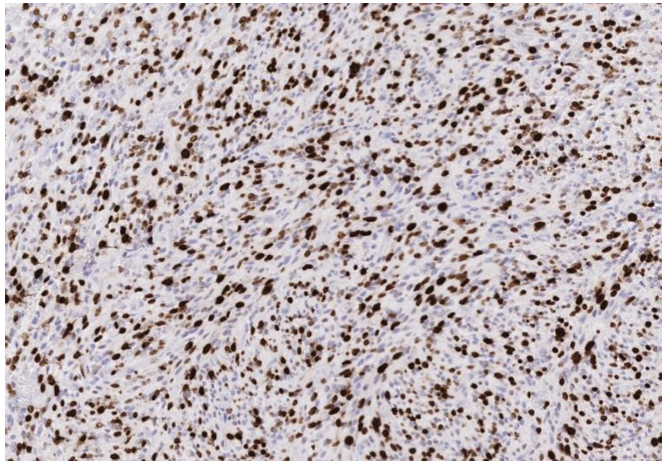


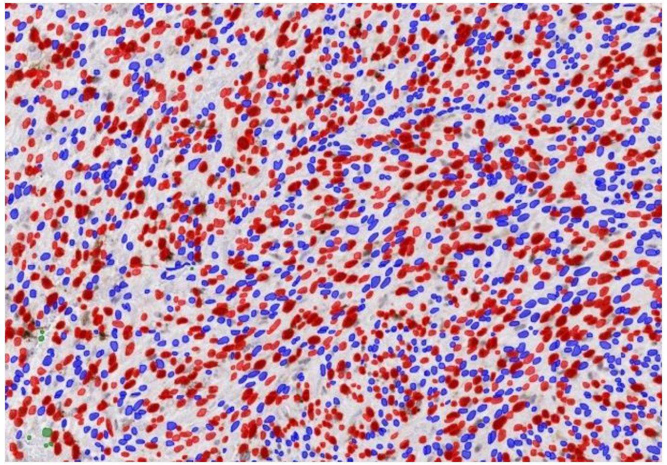


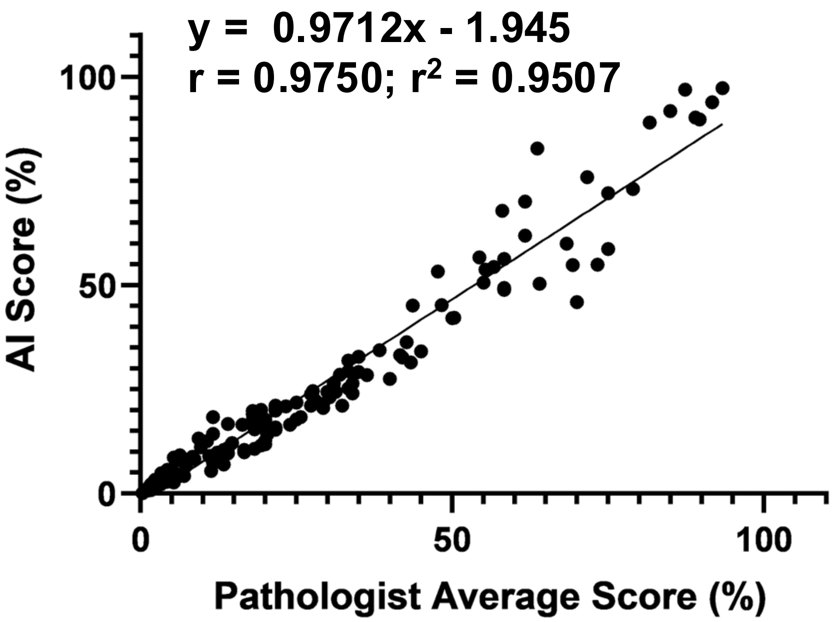


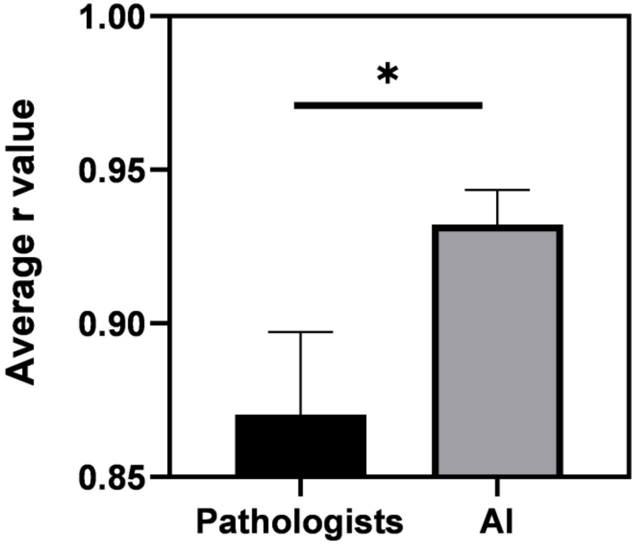
https://www.edge-ai-vision.com/2020/03/what-is-object-detection/

Cell quantification with Mask R-CNN

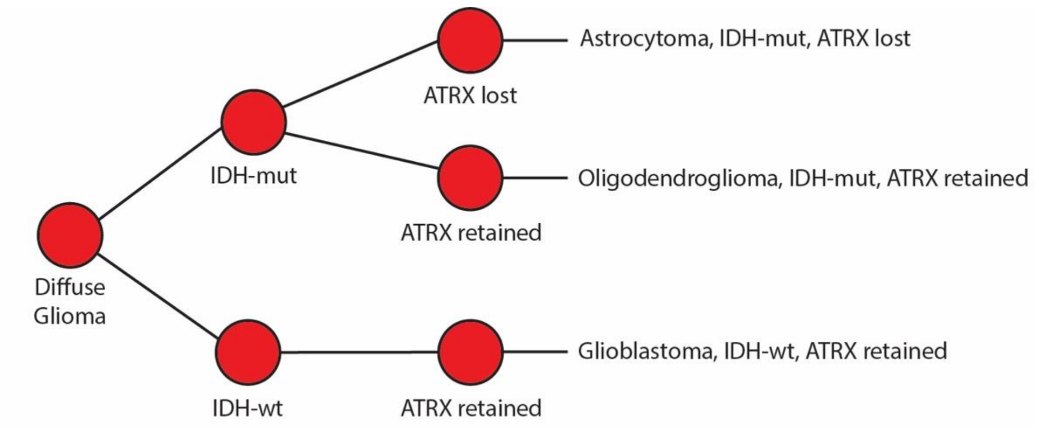


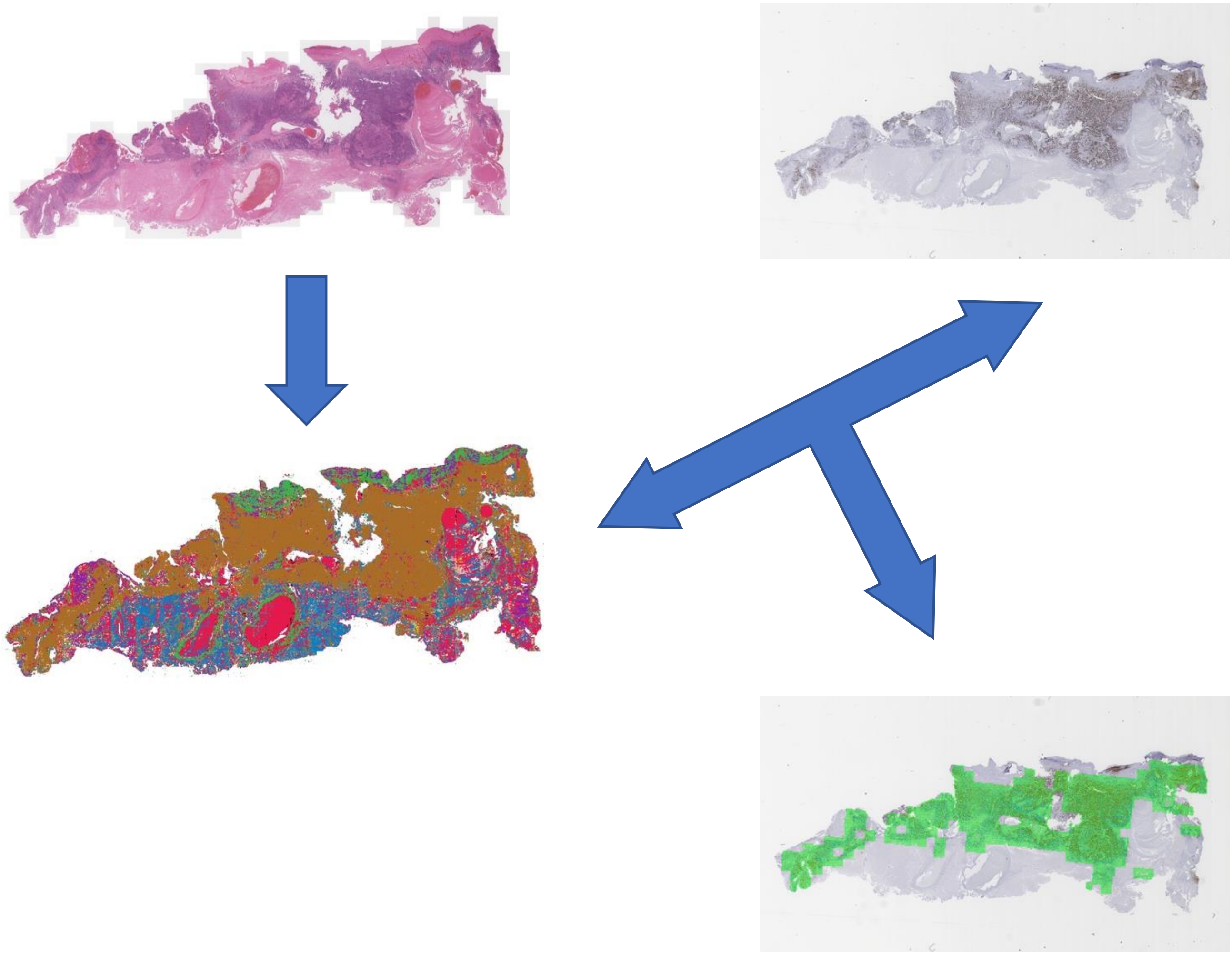






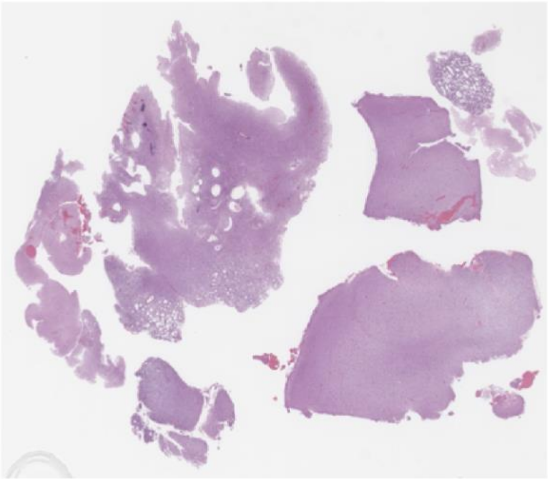
Step 3: Biomarker Interpretation (Image registration)



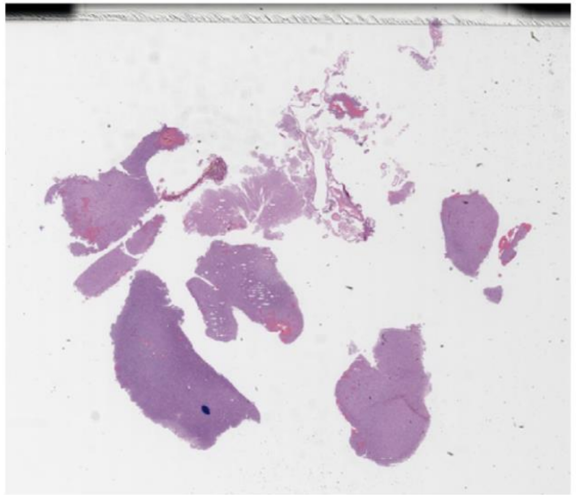


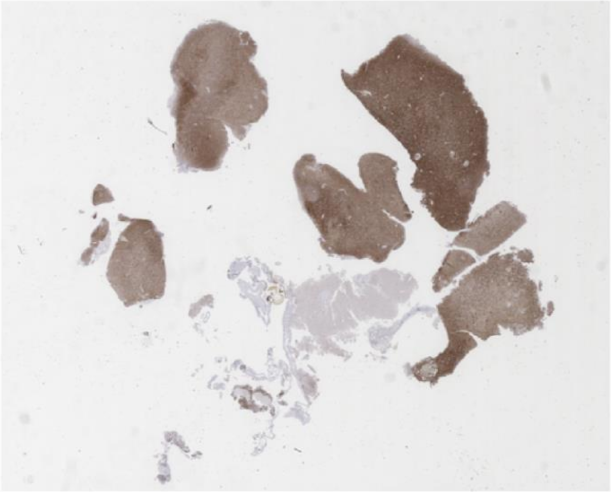
Challenges: Translation and Rotation in Tissue

Horizontal Translation

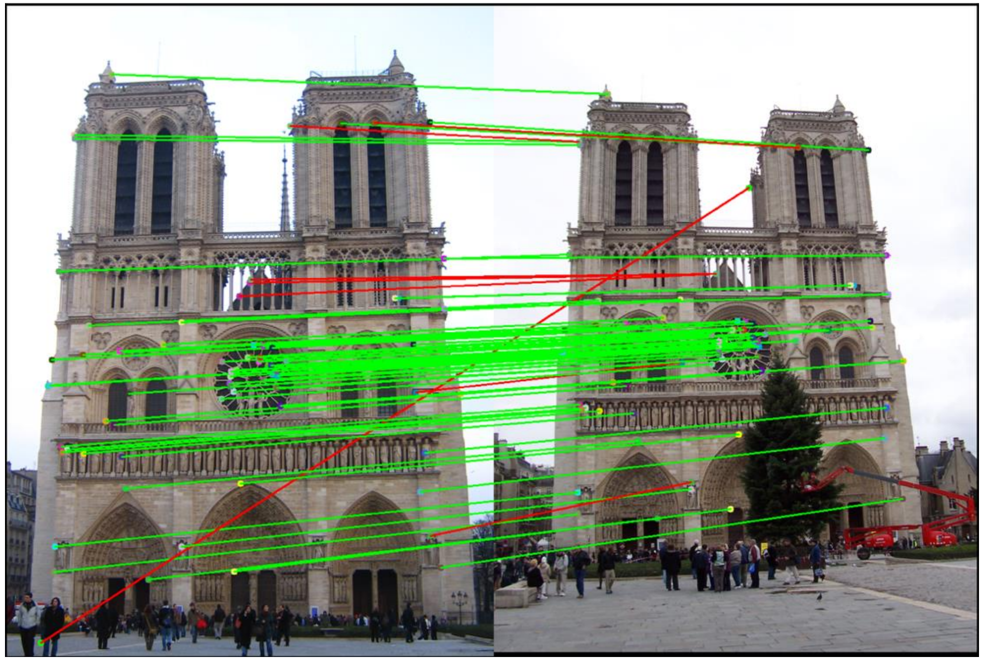


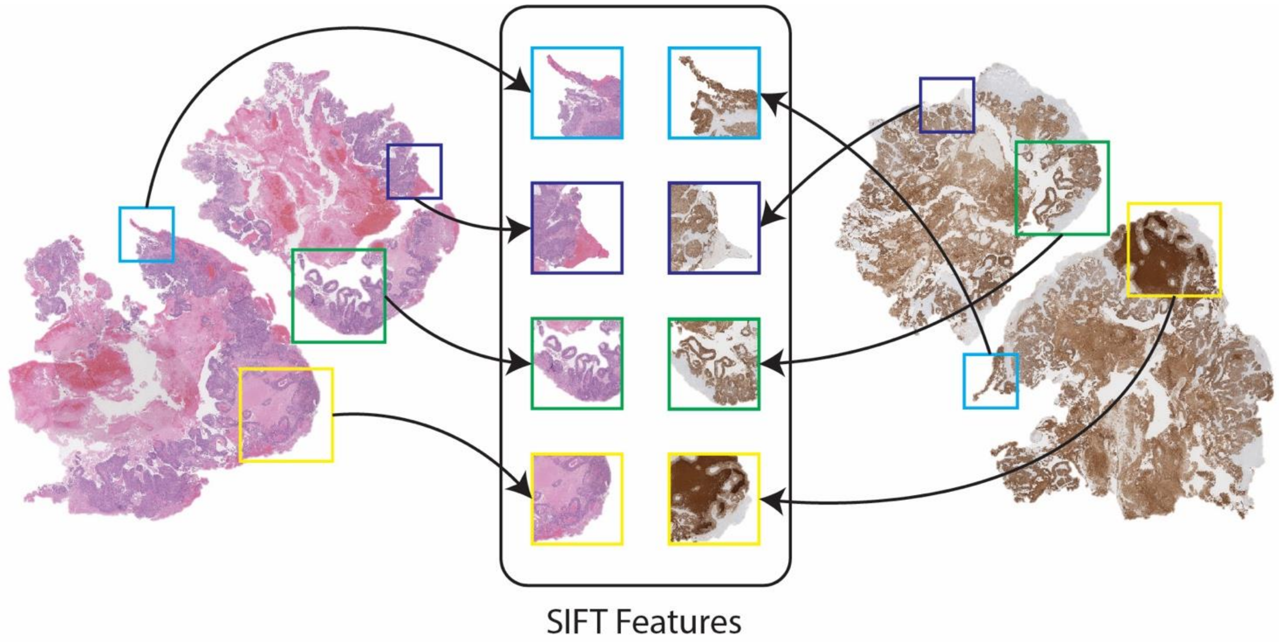
180º Rotation





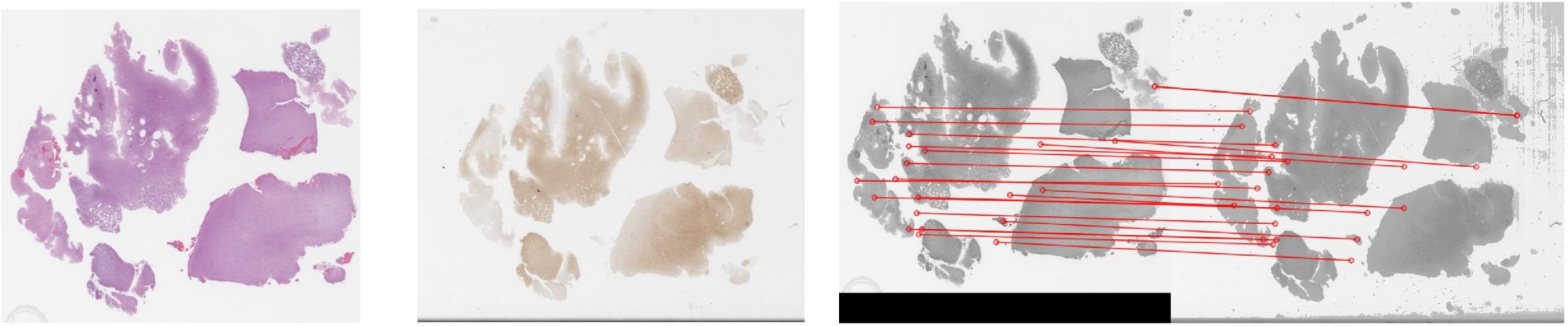
Computer vision solution: Image matching / Image Registration



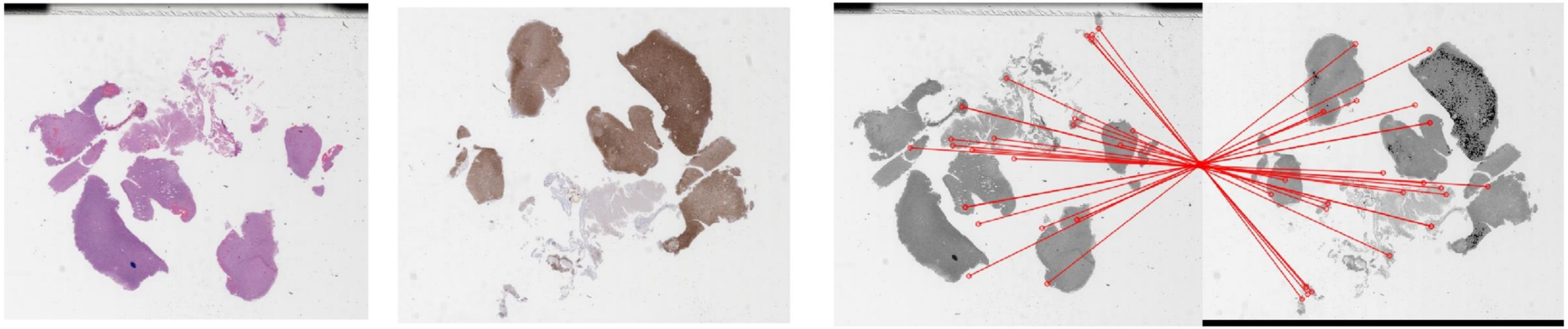


Automated WSI registration using computer vision

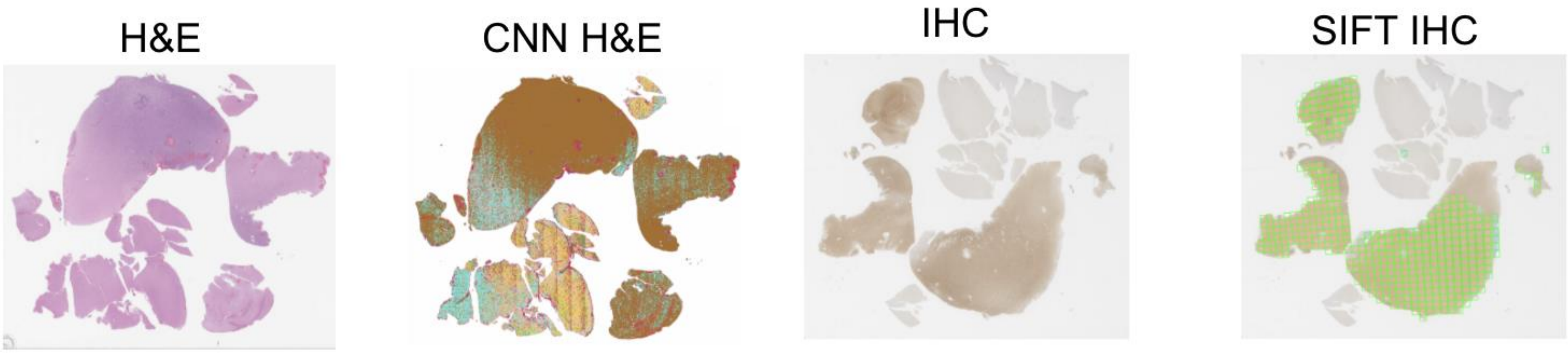
Horizontal Translation

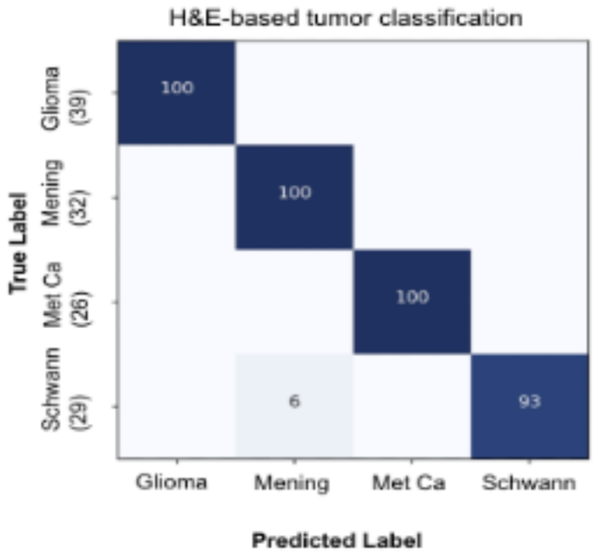


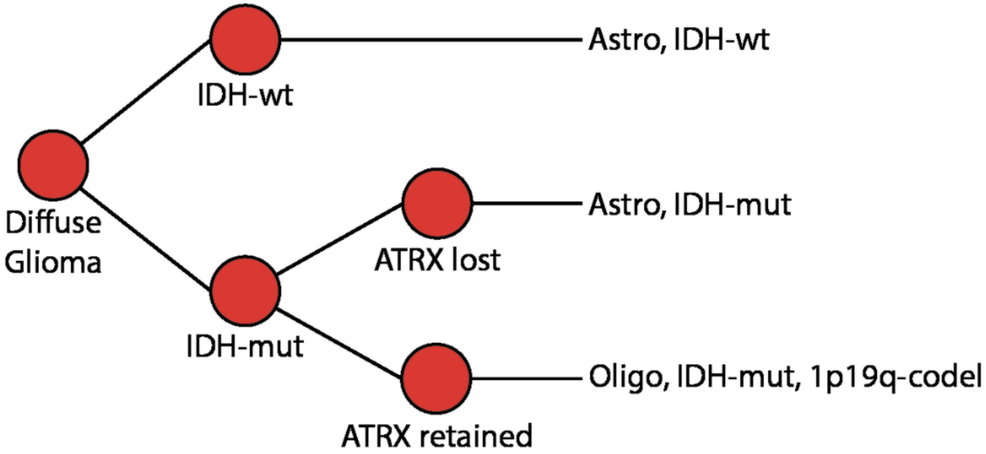
180^o Rotation

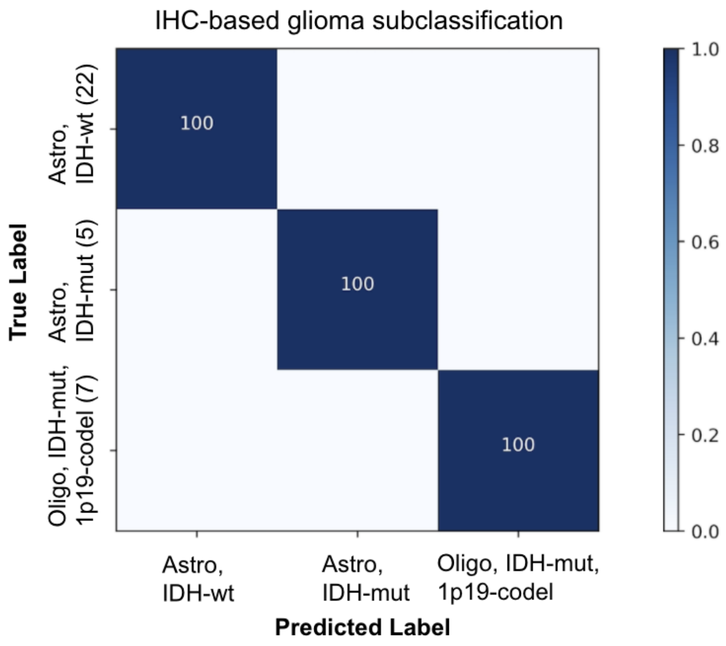


Applications: Automating molecular subclassification of brain tumors

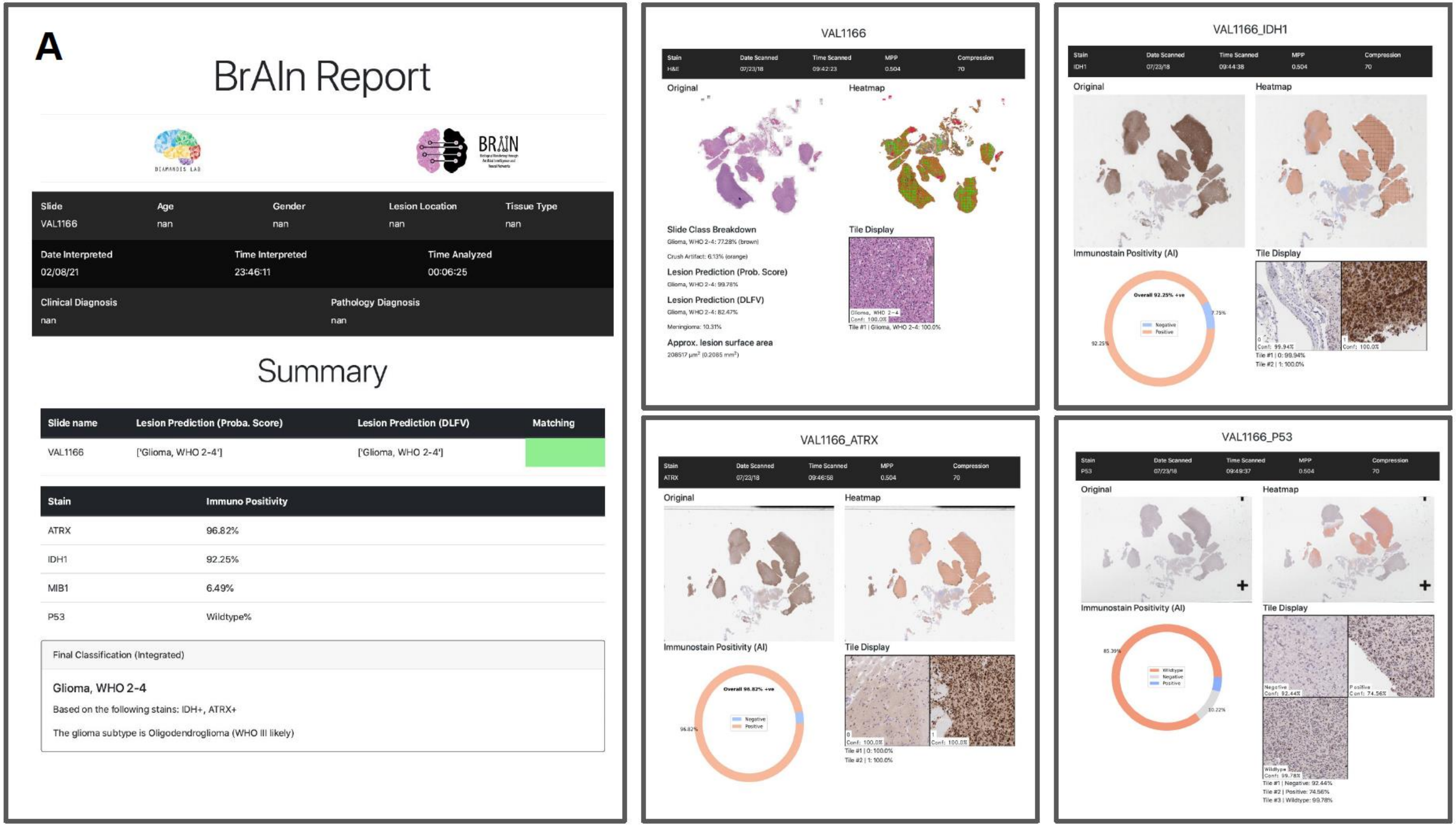








Integrating molecular information into pathology reports



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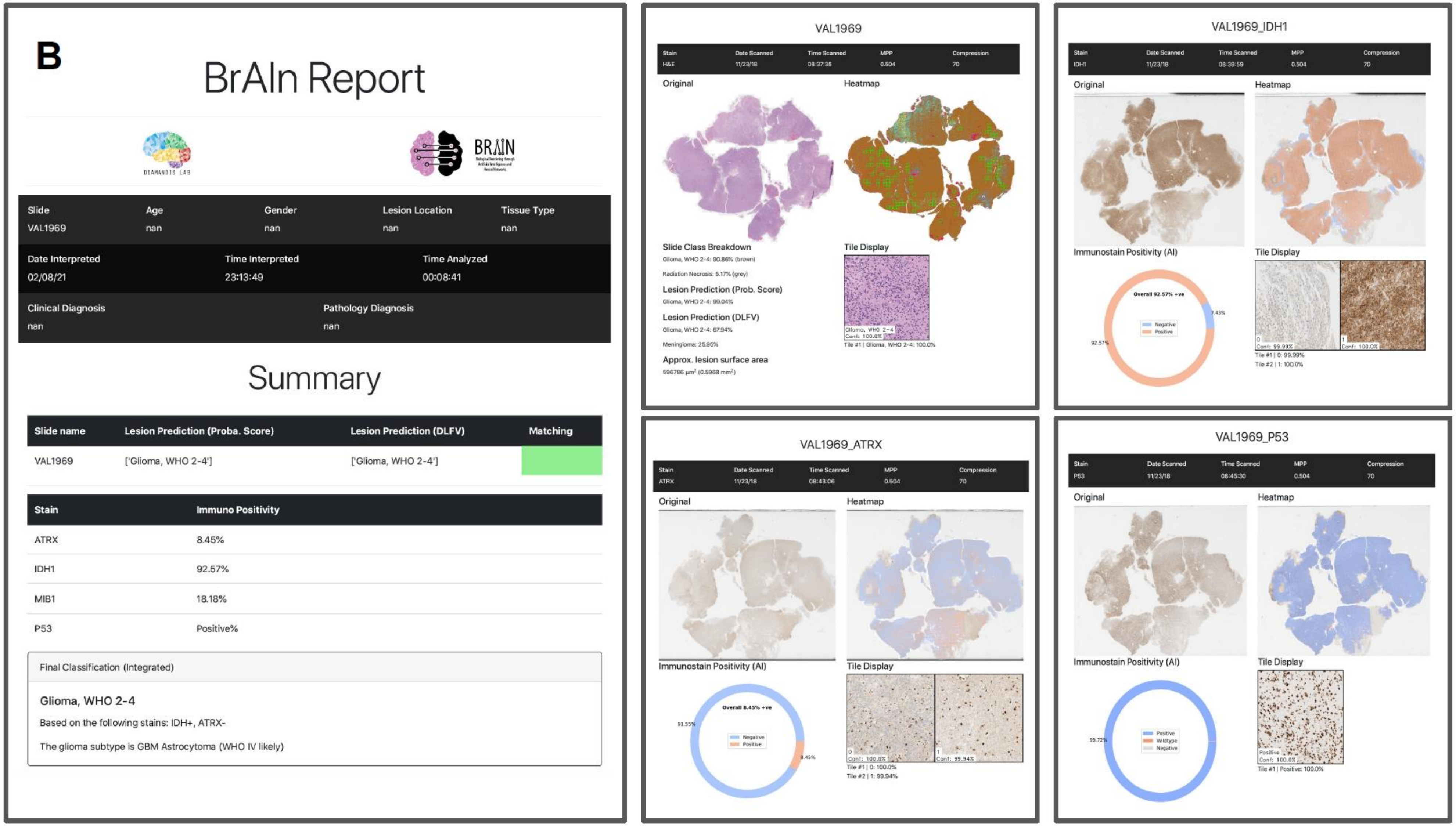
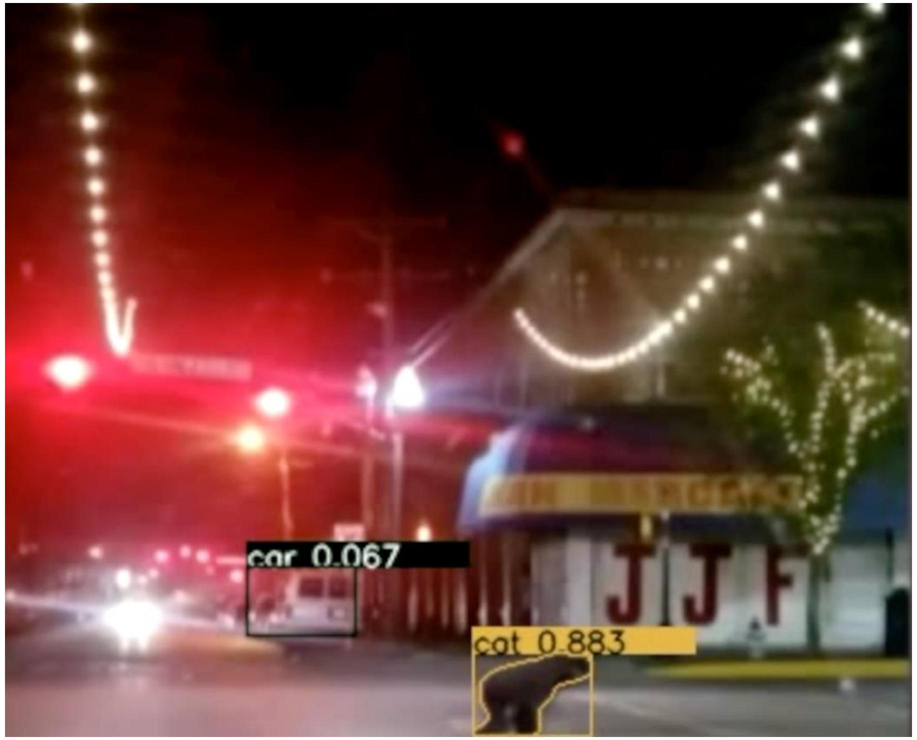


Implementation Challenges

“Partial Solutions” - Too many “edge cases”

Deep learning + Data = Not enough



Edge Cases

- "Edges": All the things that can go wrong in real world that you never expected
- Everyone logging into your site at the same time
- Hard to catalog: Individually rare, but can be collectively a common problem
- Autonomous driving: To accumulate enough rare incidents (e.g. crashes, close calls) to train vehicles, you need billions of miles → would require hundreds of years to develop training sets

From: kdnuggets.com / drisk.ai

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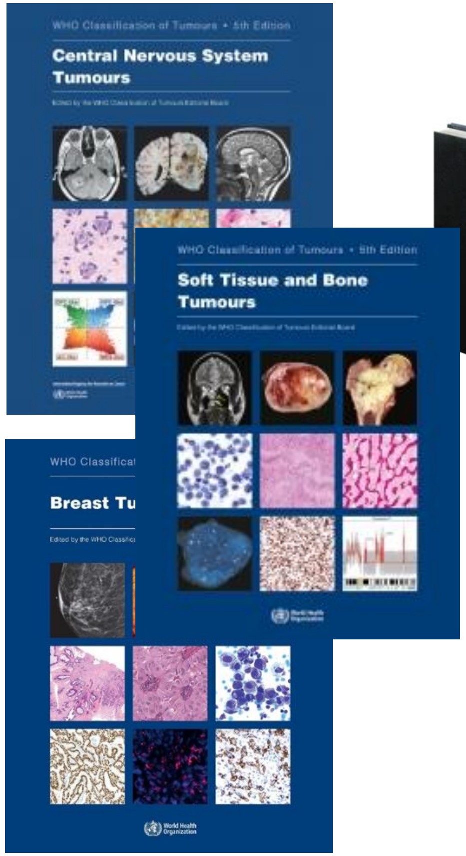
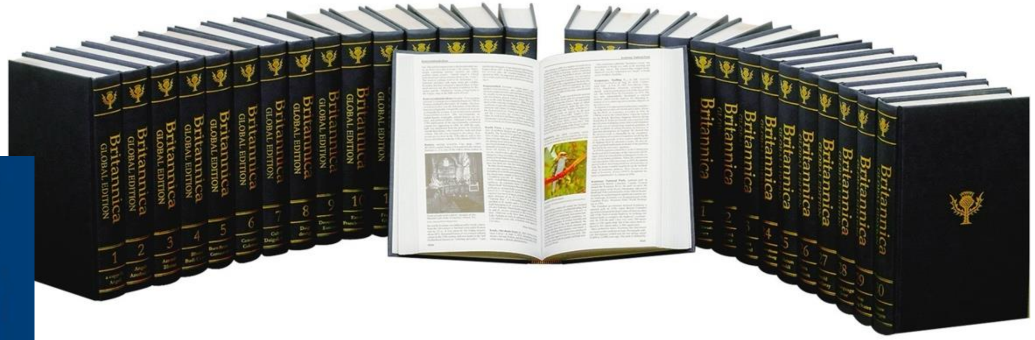
From: kdnuggets.com / drisk.ai

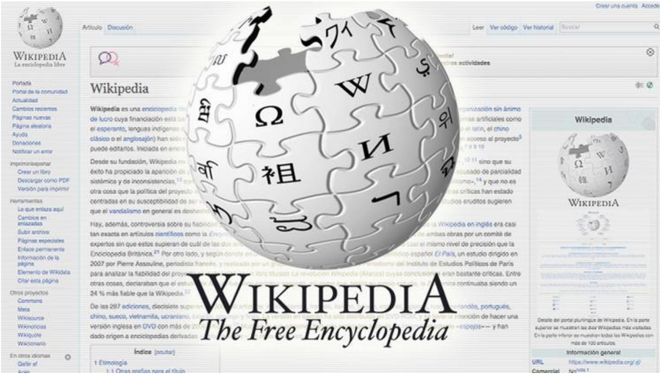
Implementation Challenges
“Partial Solutions” - Too many “edge cases”
Deep learning + Data = Not enough

- Each neural network designed for a single highly specialize task
- Each pathology subspeciality has 100’s of specialized tasks
- Multiple subspecialities

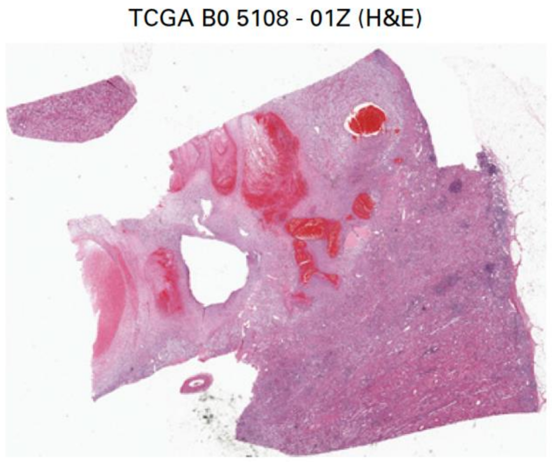
- Solution:
    - Crowdsourcing
      Computational pathology

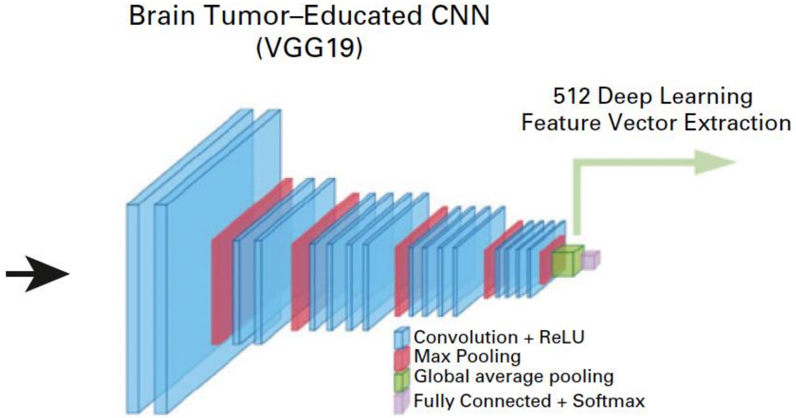


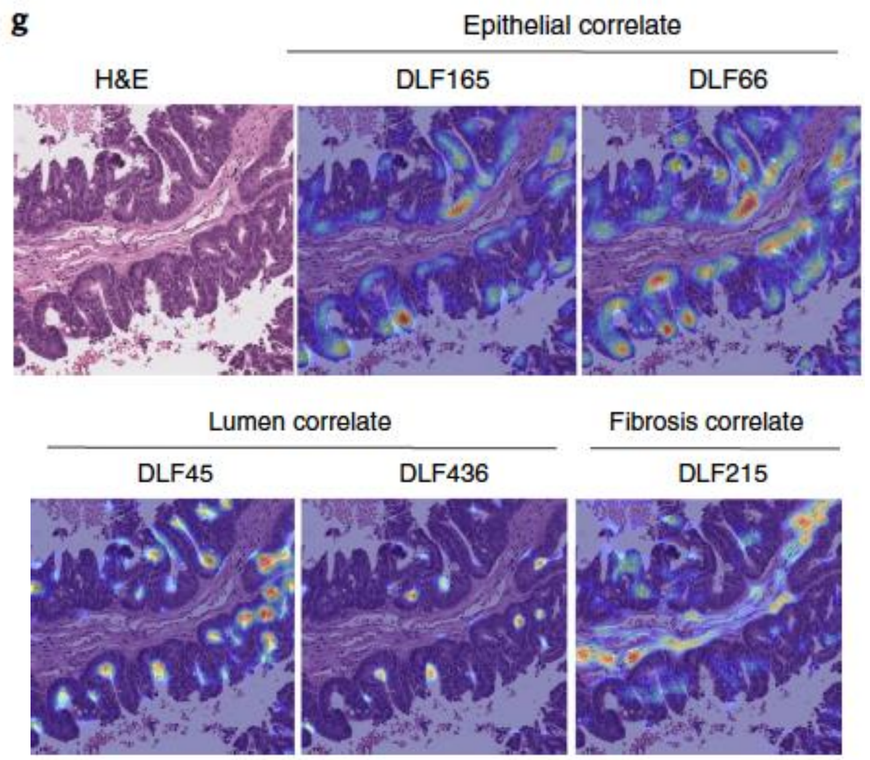




Deep Feature Extraction – Unsupervised analysis

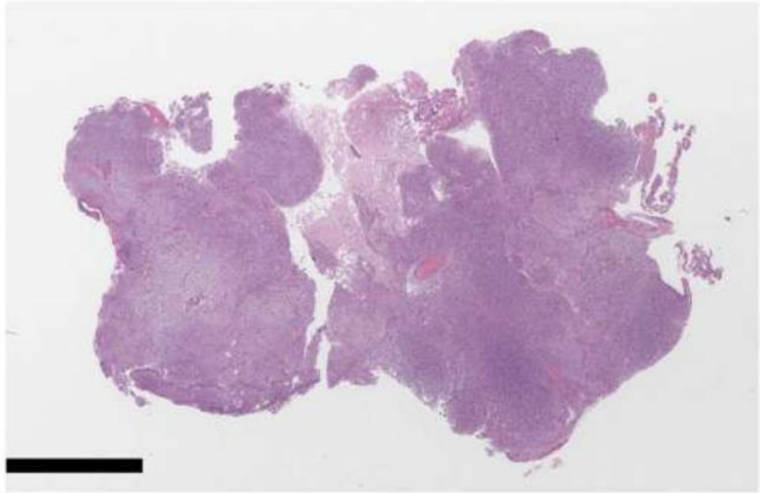


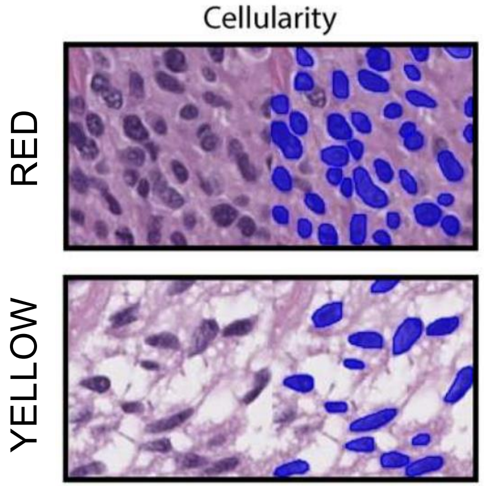


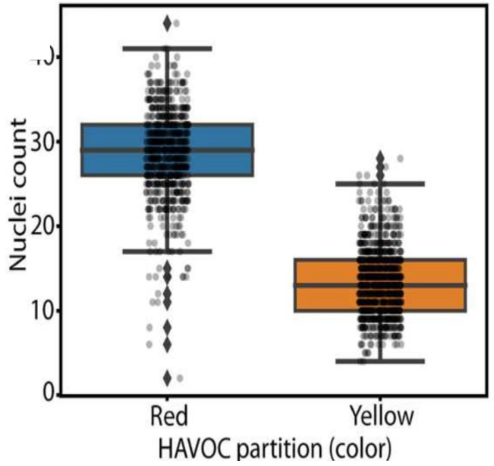


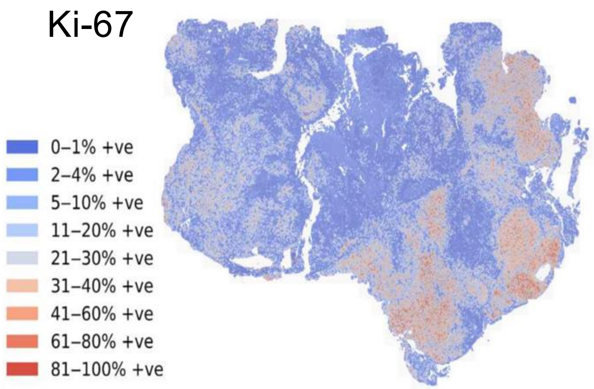
Faust et al, *Nature Machine Intelligence*, 2019

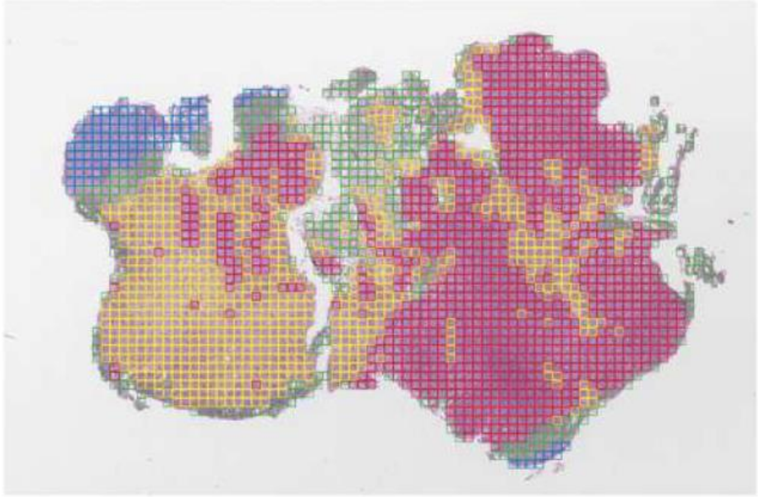
Automating analysis of Intra-Tumor Heterogeneity

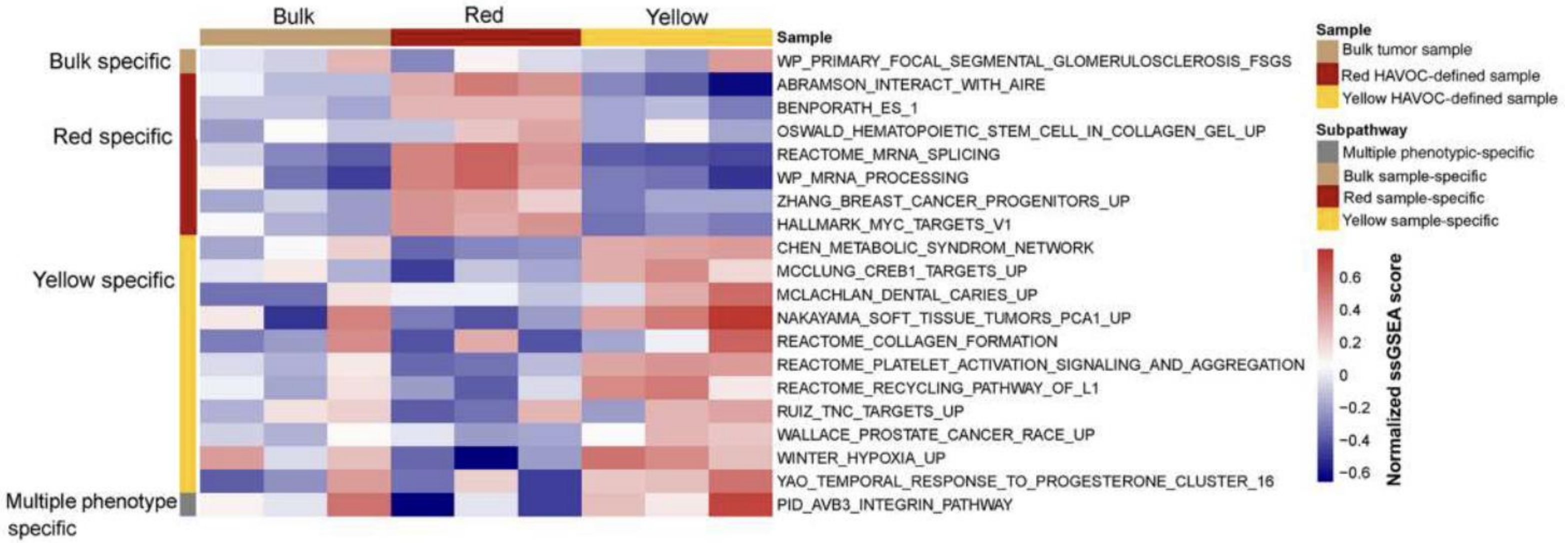






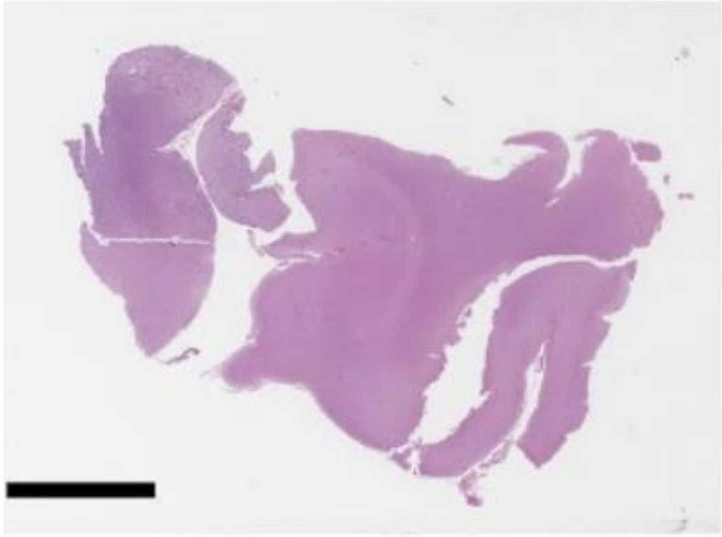


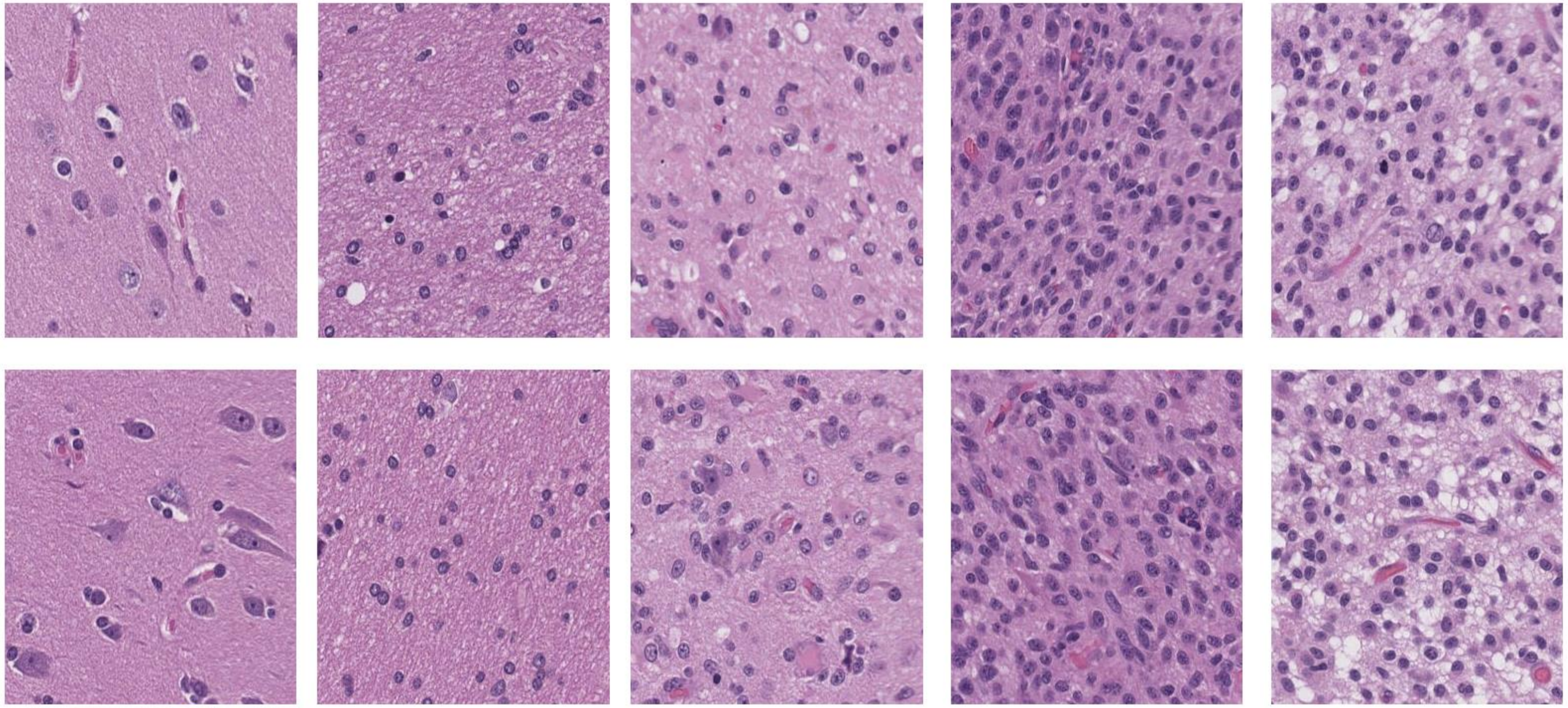




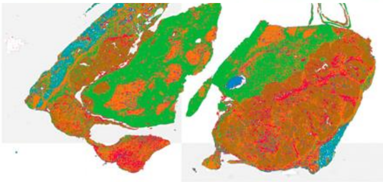
Dent & Faust, *Science Advances*, 2023

PHARAOH: Weakly supervised Active Learning + Crowd Sourcing

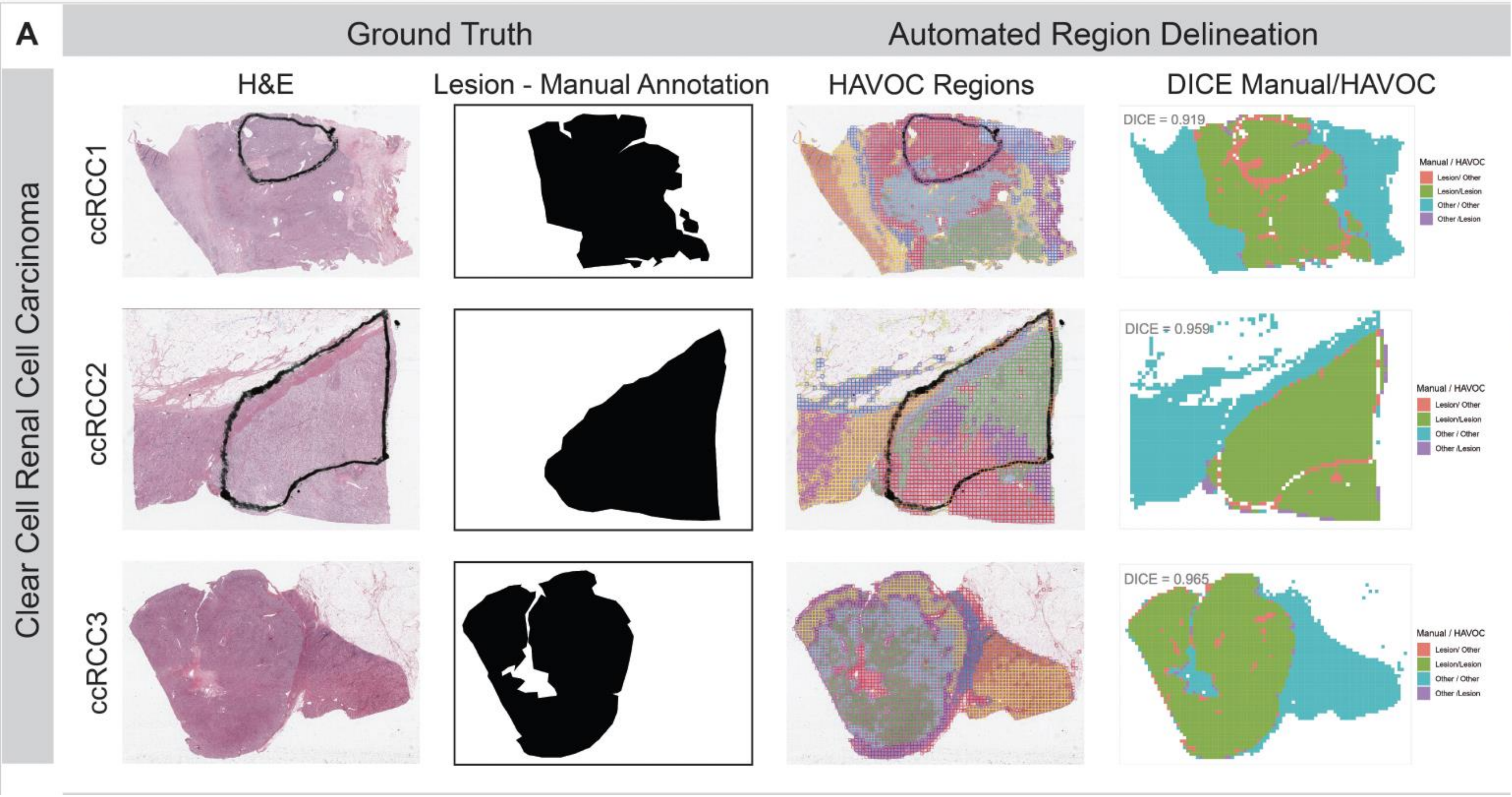




Pattern
Inferences



PHARAOH: Automated PHARAOH Segmentation and Developed Classifiers align with Expert and molecular ground truths



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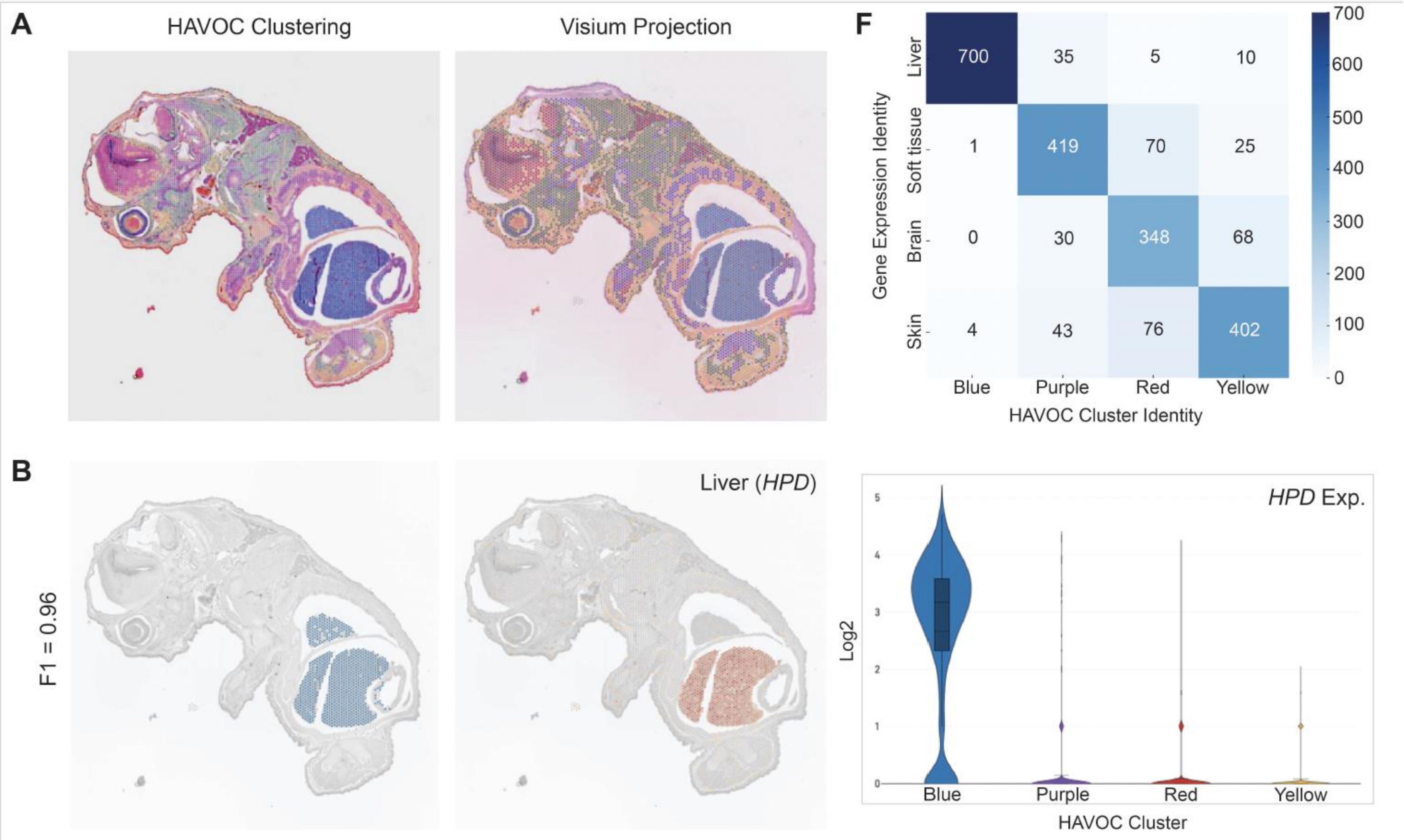

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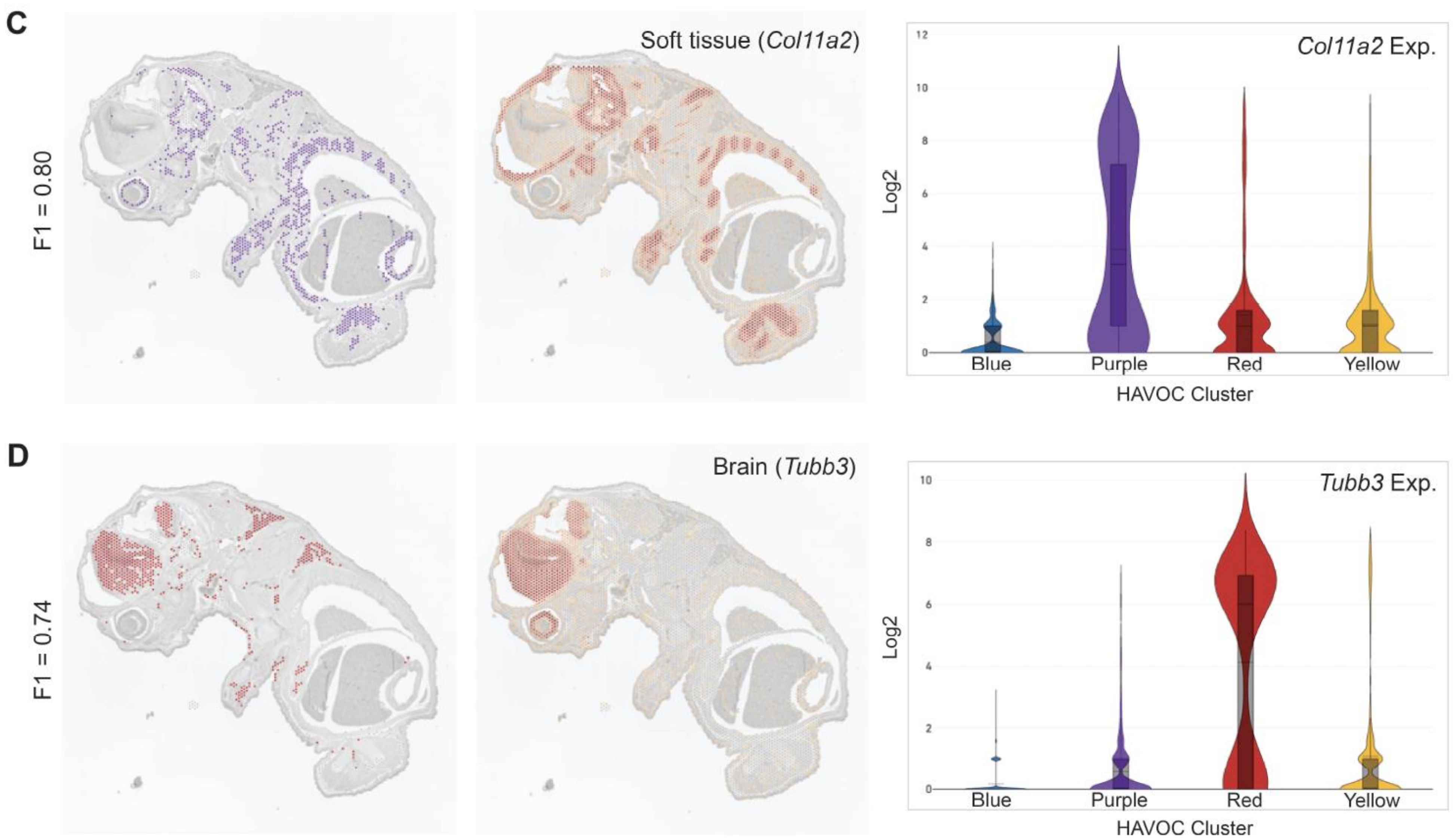
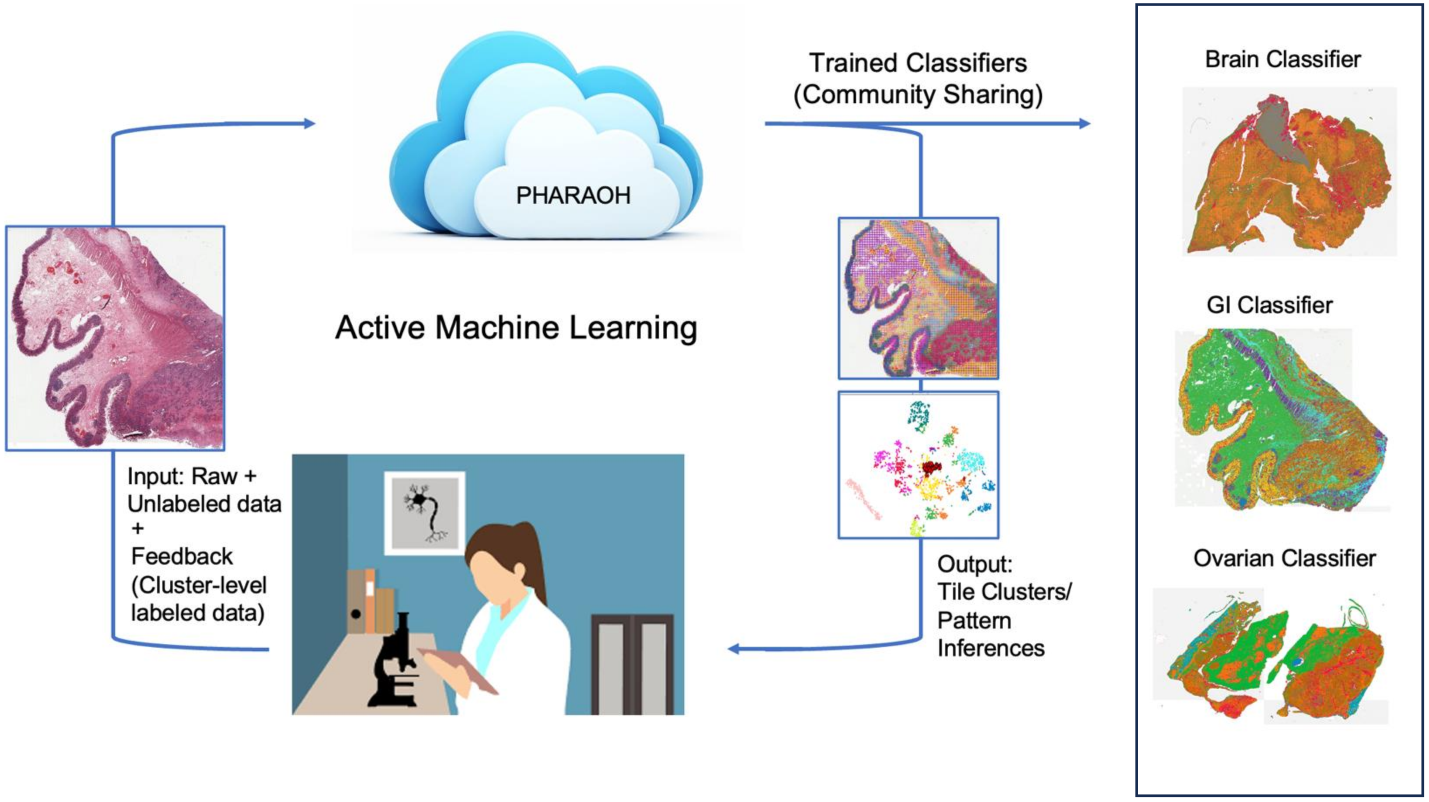
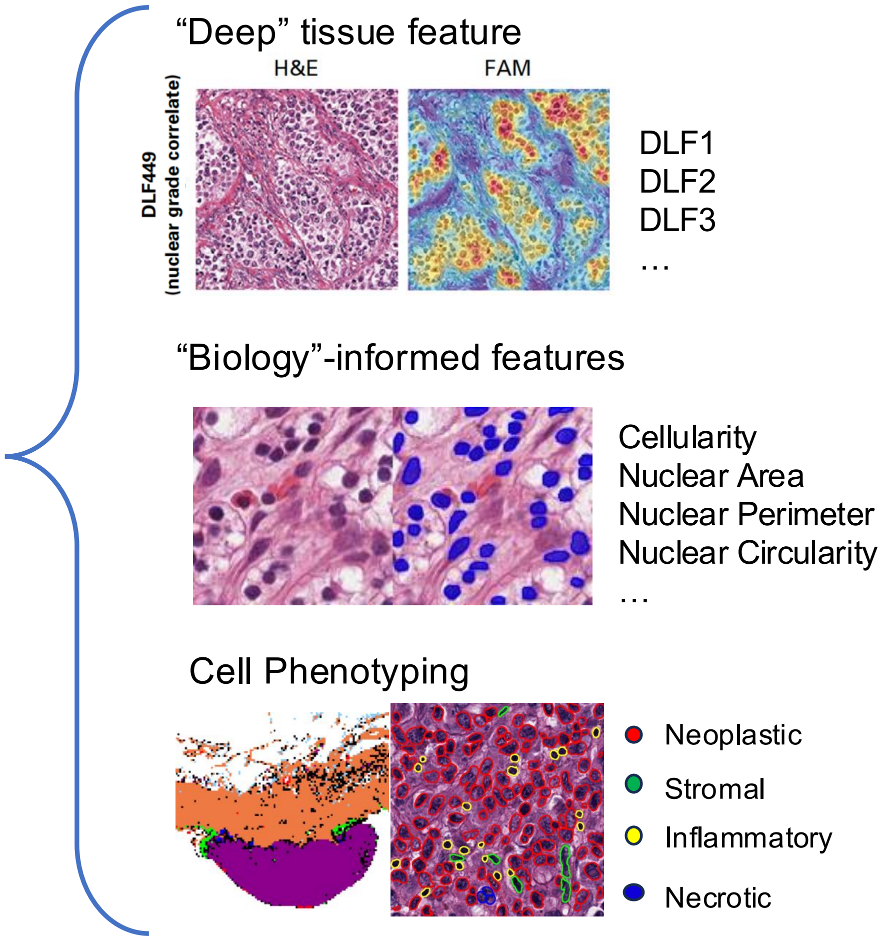


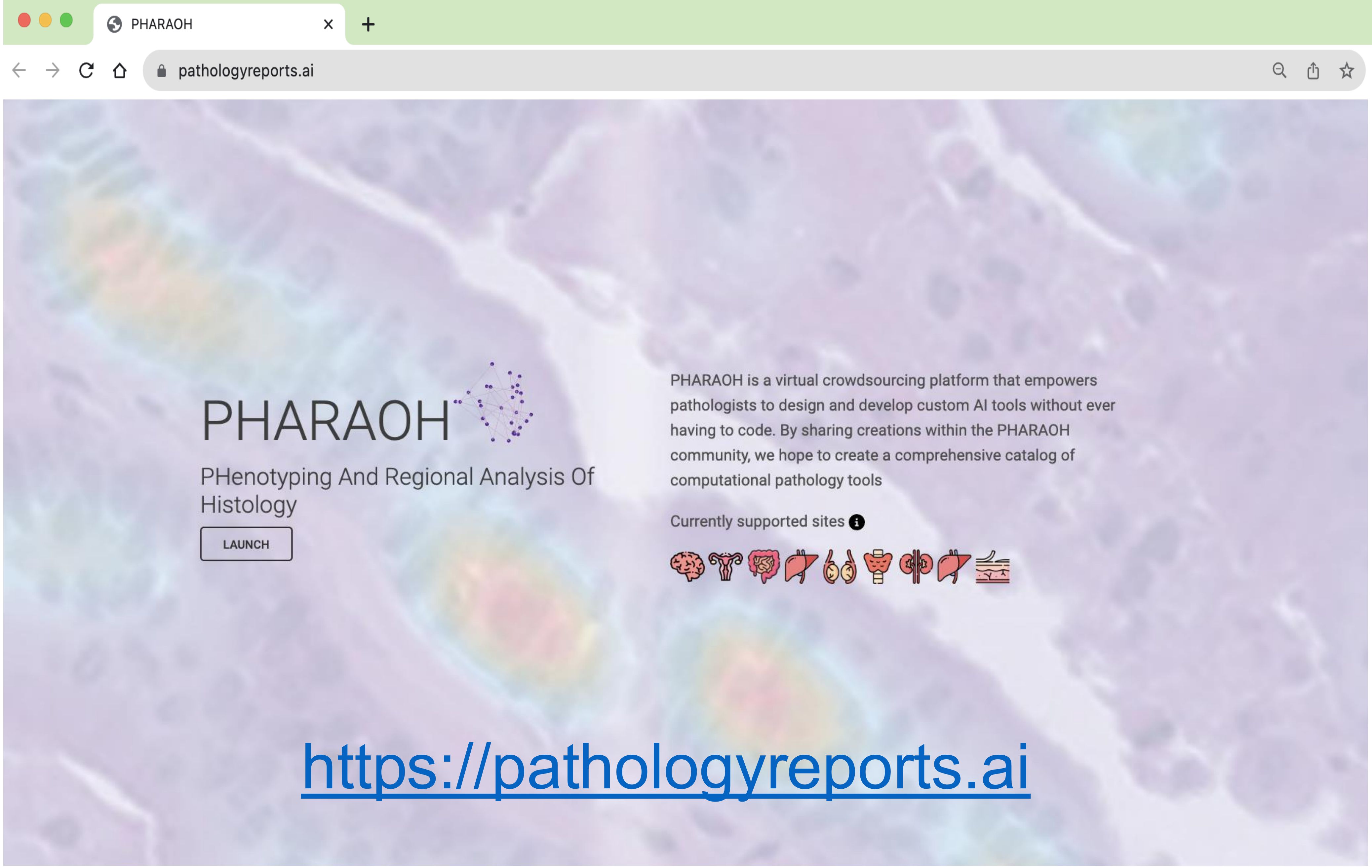
Crowdsourcing Classifier Production



Biomarker (Feature) Extraction

Outputs: Phenotyping and Regional Analysis of Histology

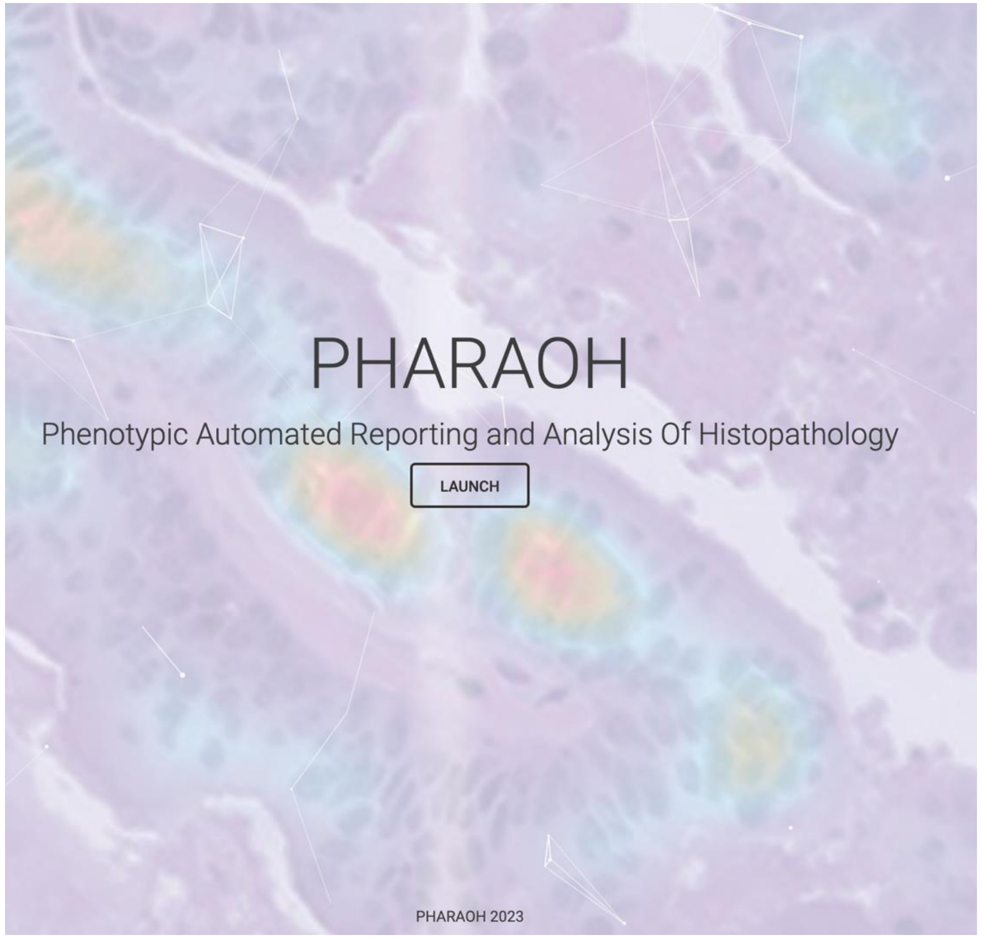




https://pathologyreports.ai

https://www.pathologyreports.ai/

PHARAOH: Crowd Sourcing Computational Pathology



Ovarian Carcinoma

Site: Ovarian

Classes:

Blank Space | Fatty Tissue | Epithelial Tumor | Fibrous Tissue | Mixture Of Epithelial Tumor And Fibrous Tissue | Necrosis | Normal Ovarian Parenchyma | Tumor edge | Hemorrhage

Description:

This H&E ovarian carcinoma tissue classifier was developed using transfer learning and the VGG19 CNN and trained to recognize ovarian serous carcinoma and other surrounding tissue elements. Annotations were carried out on batches of image tiles (dimensions: 256 x 256 um) grouped using image-based clustering (HAVOC) from 8 publicly available TCGA-OV H&E-stained whole slide images. Validation testing was carried out on non-overlapping image tiles from the same cases.

***Designed by:** Dr. Noor Alsafwani (King Fahd Hospital of University, Saudi Arabia)*

SELECT

Gastrointestinal Adenocarcinoma

Site: GI

Classes:

Connective tissue | Edge of Tumor | Edge Of Tumor And Inflammtory Cells | Epithelial Pattern | Neoplastic epithelial pattern | Inflammatory Cells | Mucin | Smooth muscle | Acute hemorrhage | Blank space | Necrosis

Description:

This H&E colorectal carcinoma tissue classifier was developed using transfer learning and the VGG19 CNN and trained to recognize colorectal adenocarcinoma and other surrounding tissue elements. Annotations were carried out on batches of image tiles (dimensions: 256 x 256 um) grouped using image-based clustering (HAVOC) from 8 publicly available TCGA-COAD H&E-stained whole slide images. Validation testing was carried out on non-overlapping image tiles from the same cases.

***Designed by:** Dr. Mohamed Al-Yousef (King Fahd Hospital of University, Saudi Arabia)*

SELECT

Common Brain Tumors and Surrounding Tissue Elements

Site: Brain

Classes:

Acute hemorrhage | Blank space | Cerebellar cortex | Chronic hemorrhage | Crush Artifact | Fibrocollagenous tissue | Glial histology | Gray matter | Normal hematolymphoid tissue | Meningothelial Histology | Epithelial pattern | Skeletal muscle | Necrosis | Radiation Necrosis | Schwannian/spindled Histology | White matter

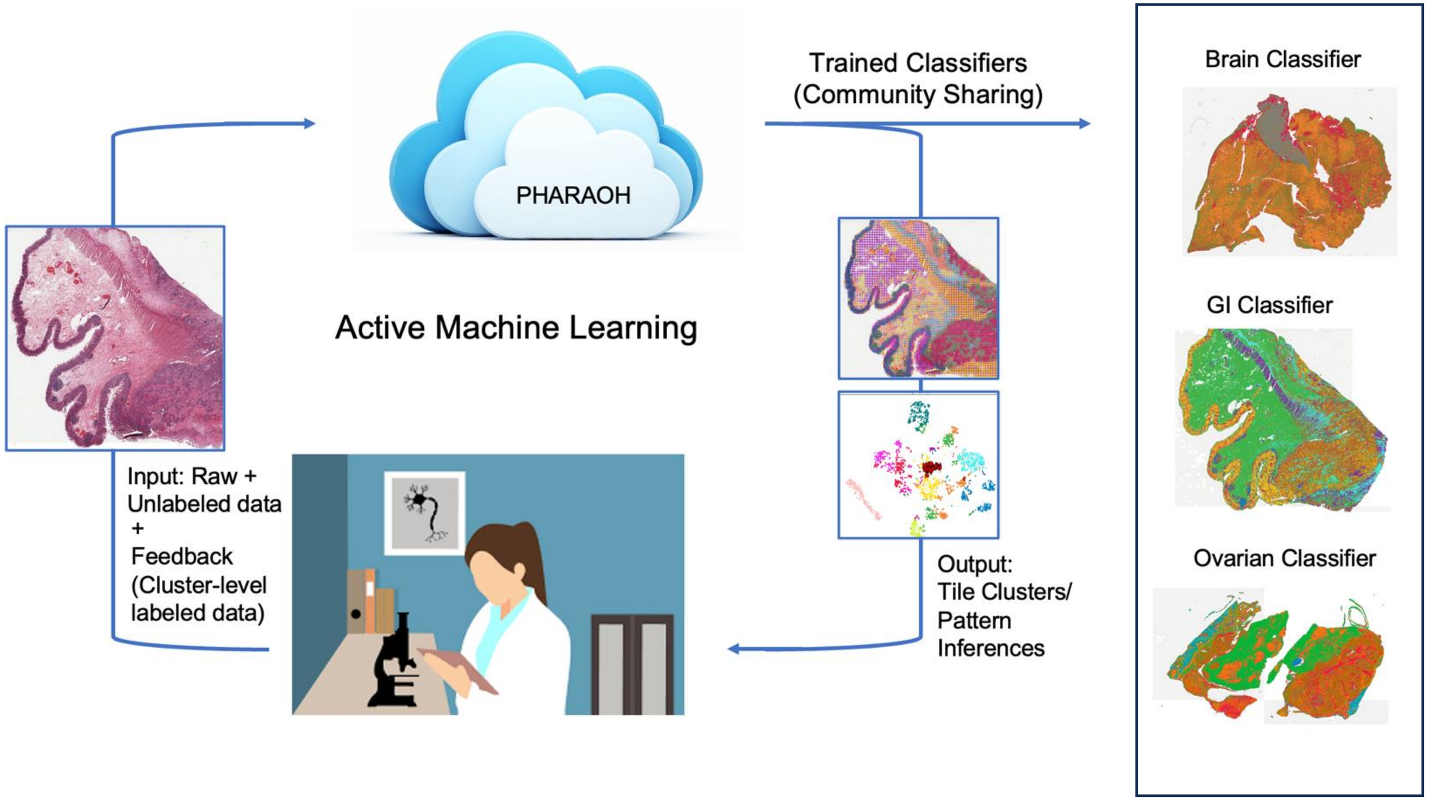
Description:

This H&E brain tumor classifier classifier was developed using transfer learning and the VGG19 CNN and trained to recognize the four most common brain tumor types and other surrounding tissue elements. H&E-stained images patches (dimensions: 512 x 512 um) and annotations were taken from previous publicly available datasets (PMID: 35156037). Validation testing was carried out on non-overlapping image tiles from independent cases.

***Designed by:** Kevin Faust & Michael Lee and Colleagues (University Health Network, Canada)*

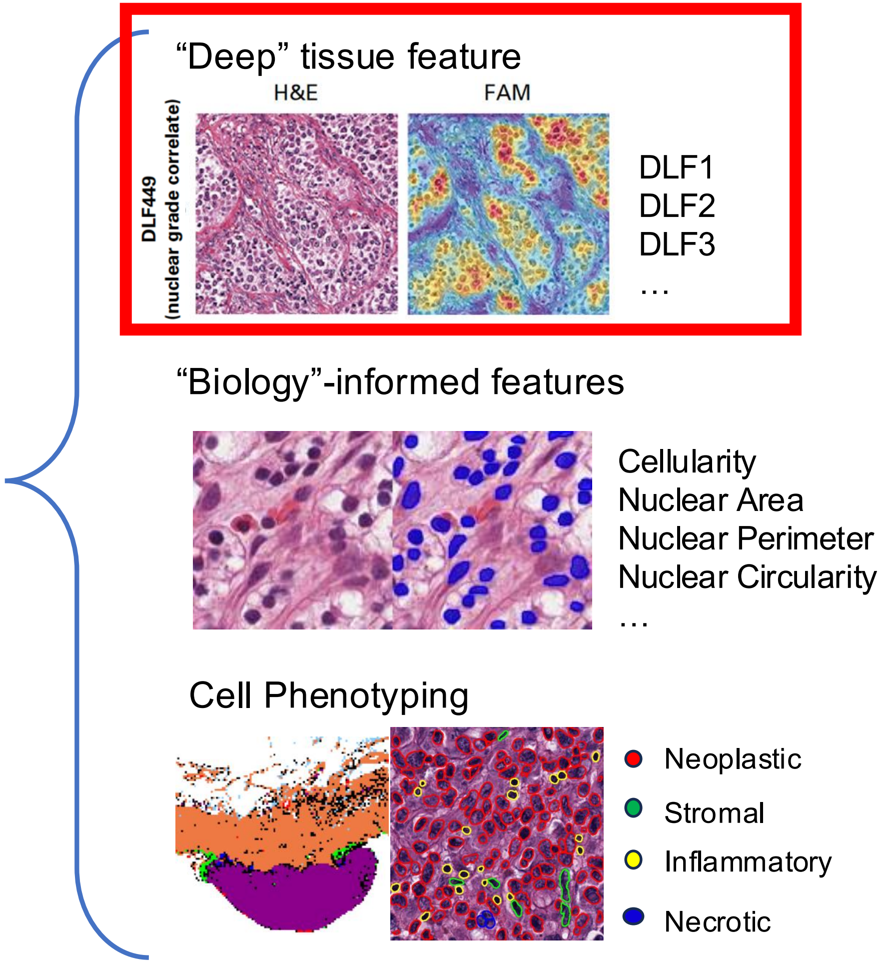
SELECT

Crowdsourcing Classifier Production

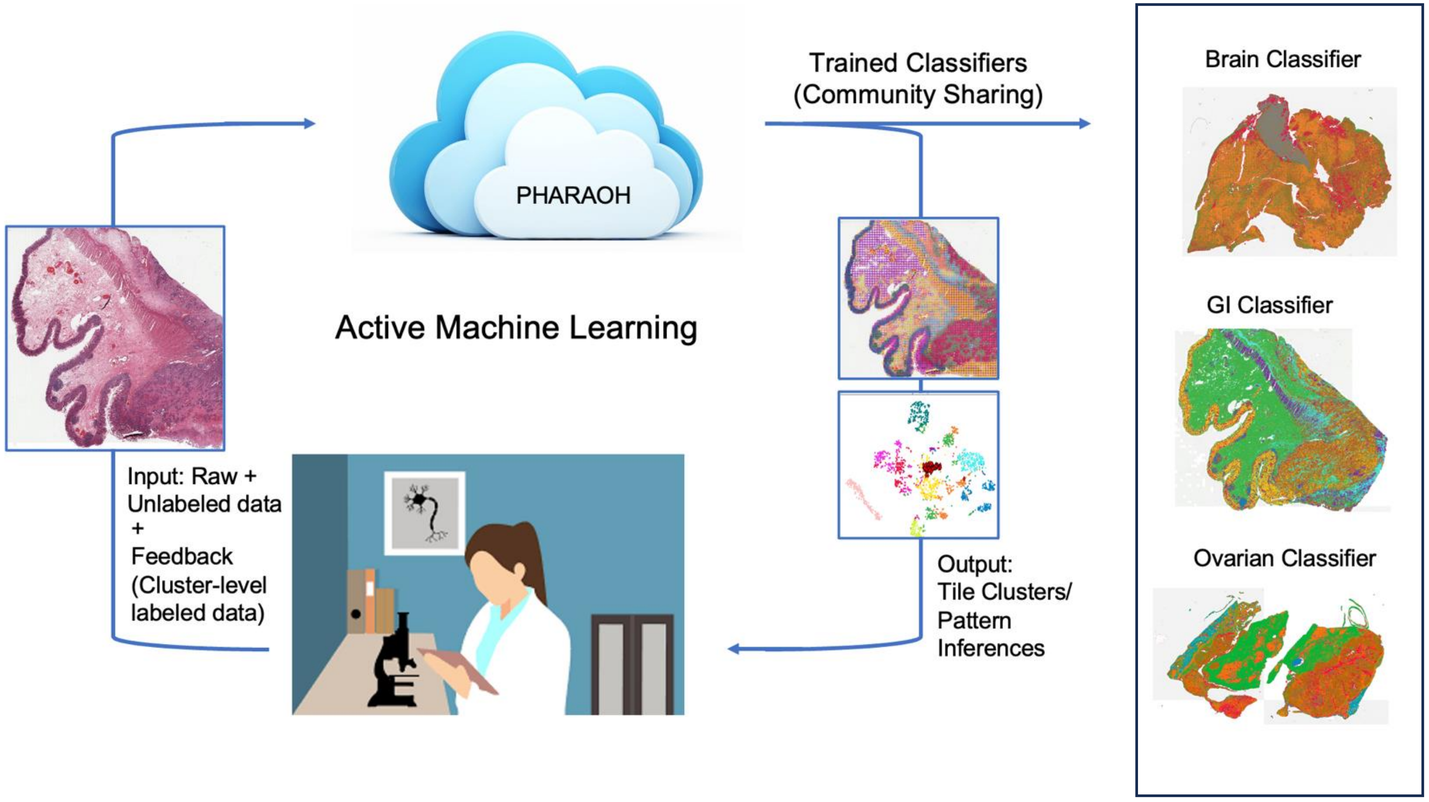


Biomarker (Feature) extraction

Outputs: Phenotyping and Regional Analysis of Histology

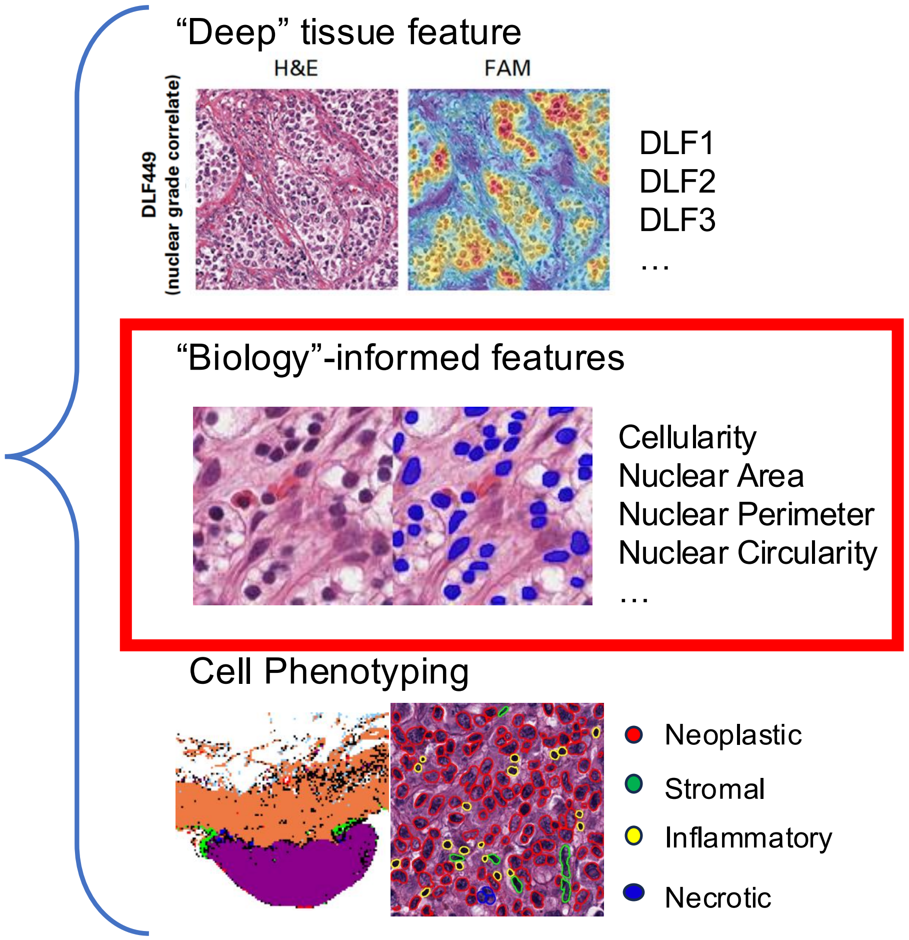


Crowdsourcing Classifier Production

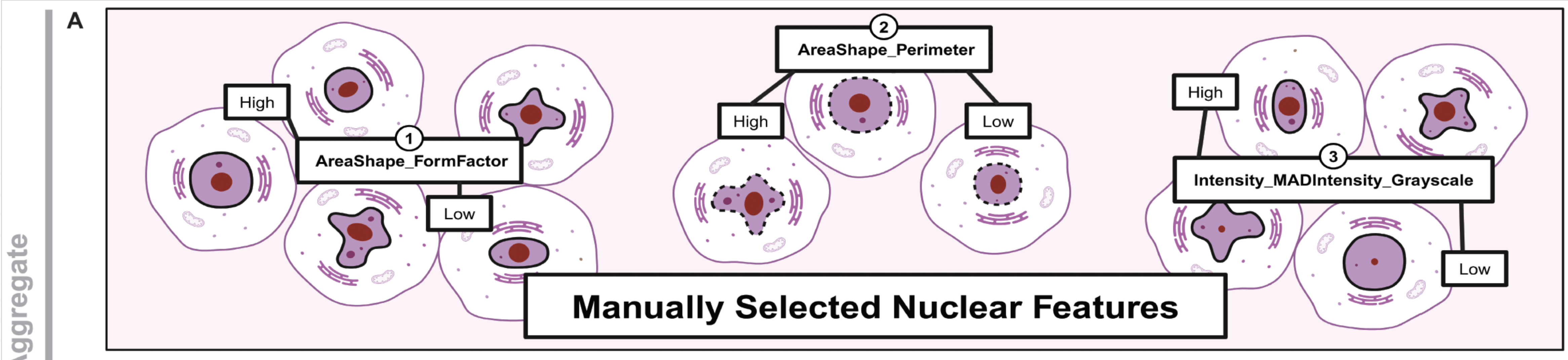


Biomarker (Feature) extraction

Outputs: Phenotyping and Regional Analysis of Histology

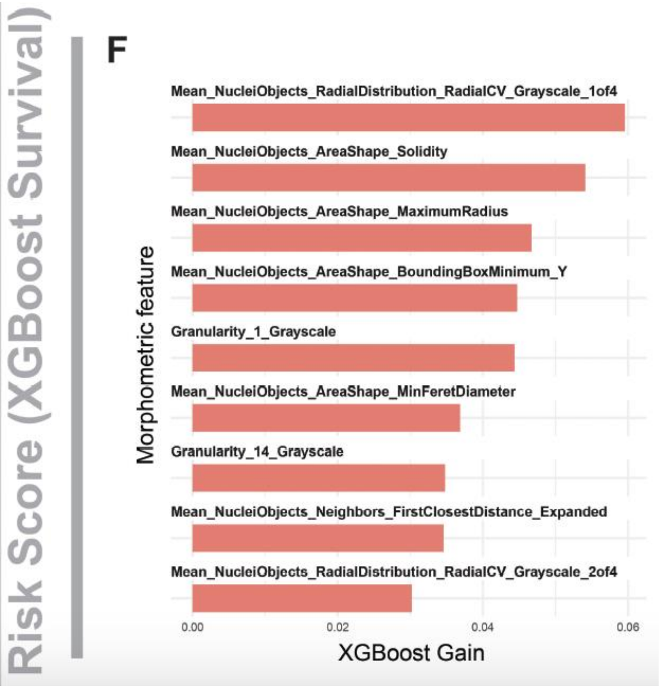


PHARAOH Applications – Tumor Grade automation

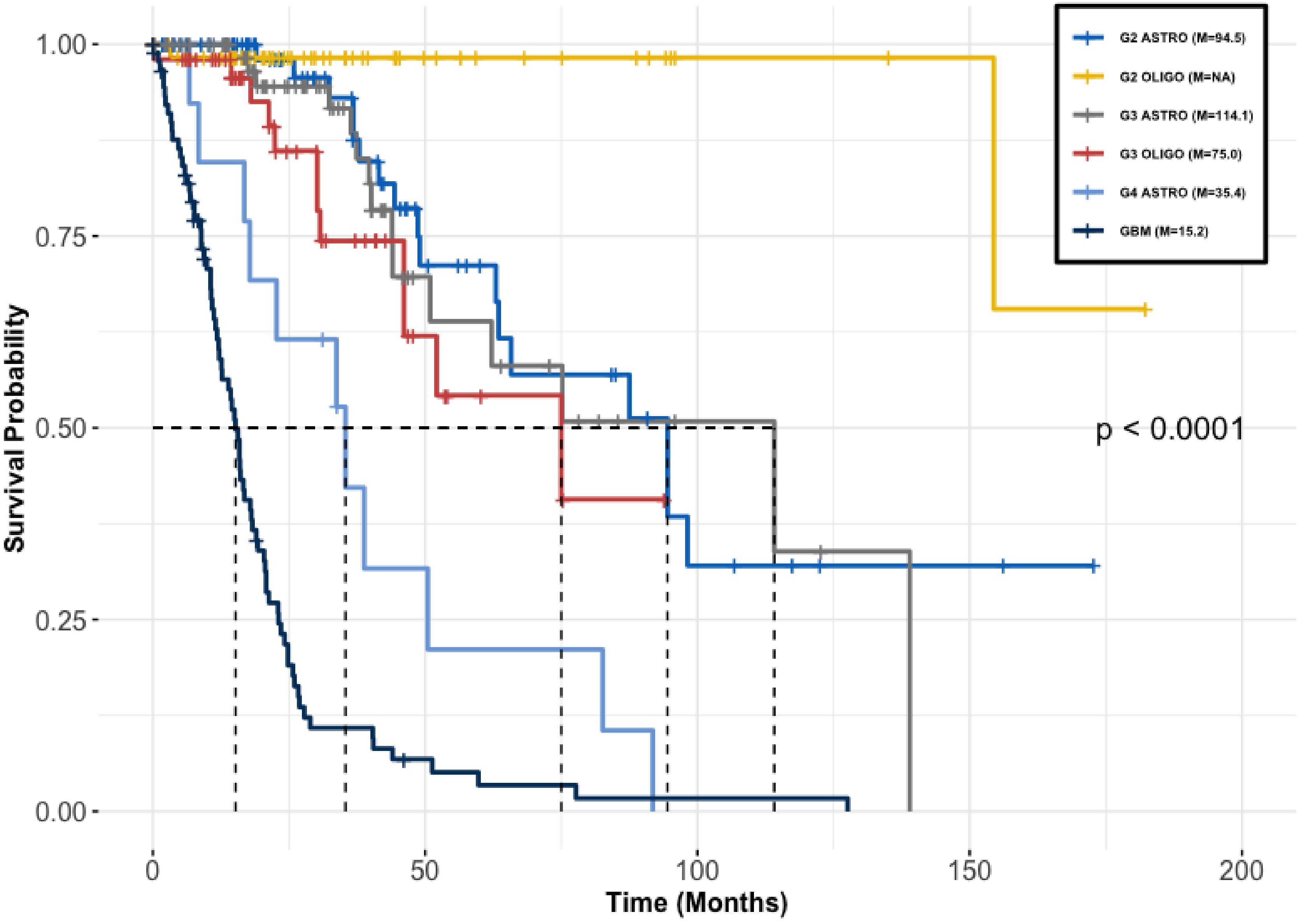


Grade	Nuclei		Nucleoli
	Size (µm)	Shape	
1	10	Round, Uniform	Inconspicious or absent
2	15	Slightly irregular	Evident at high power (X400 magnification)
3	20	Obviously irregular	Prominent, large at low power (X100 magnification)
4	>20	Bizzare, Often multilobed	Heavy chromatin clump

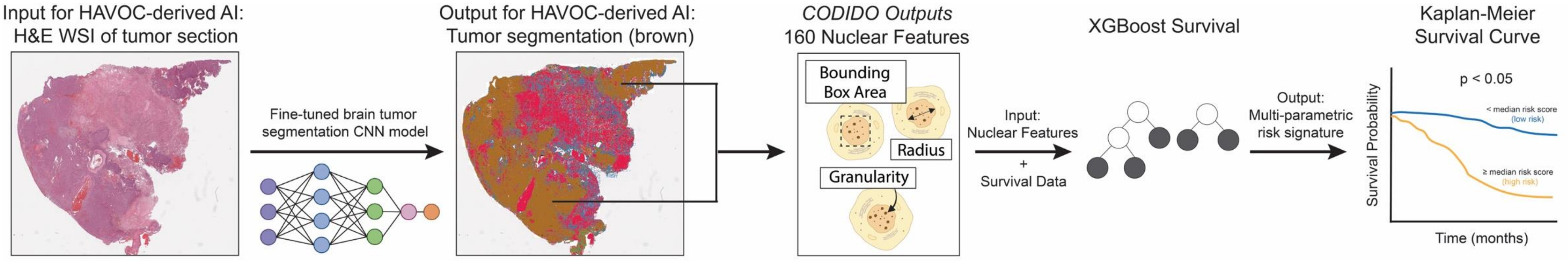
PHARAOH Applications – Data driven risk score design

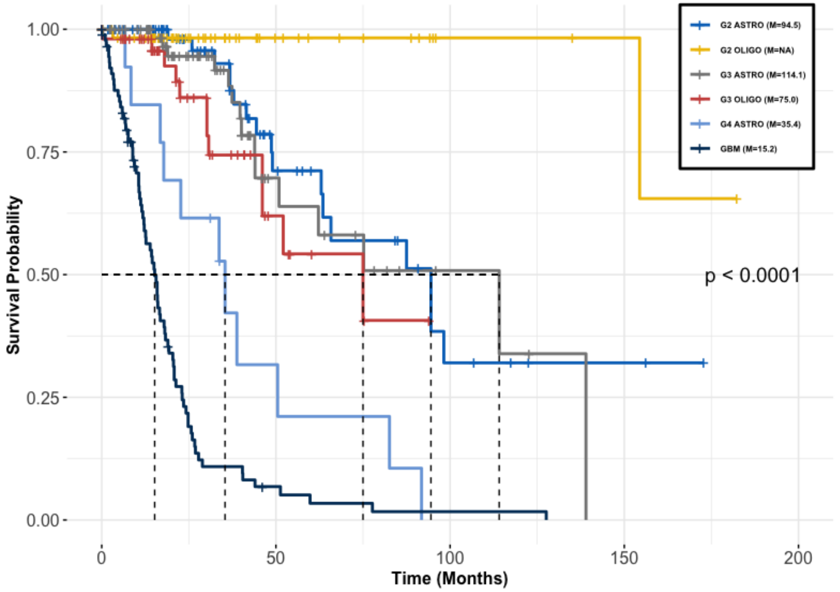


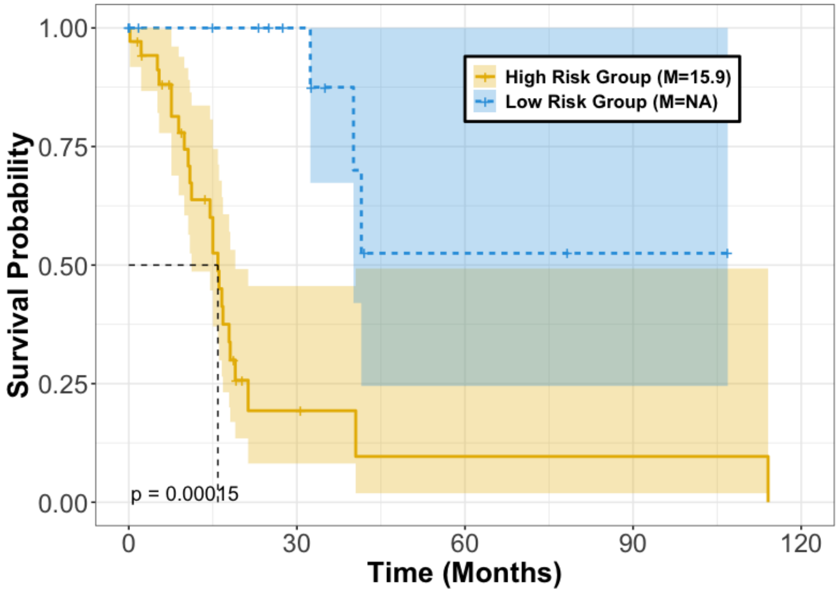
Risk Prediction: IDH-mutated 1p19q codel Oligodendrogliomas, WHO Gr 3



Risk Prediction: IDH-mutated 1p19q codel Oligodendrogliomas, WHO Gr 3







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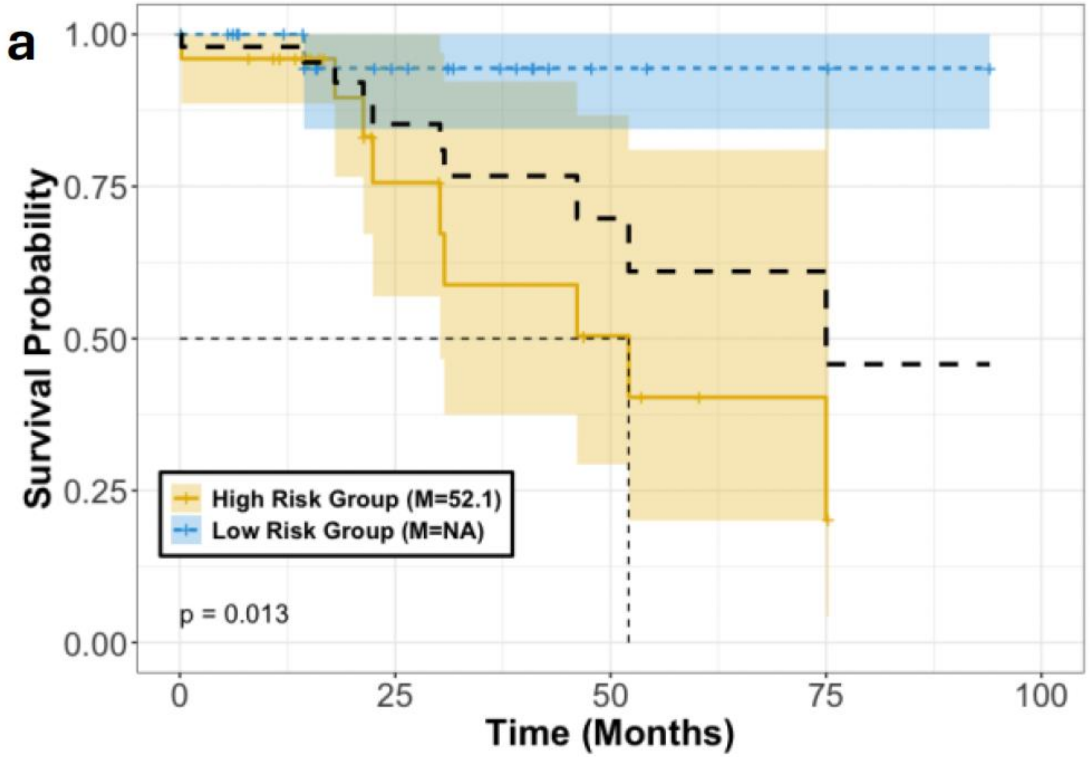

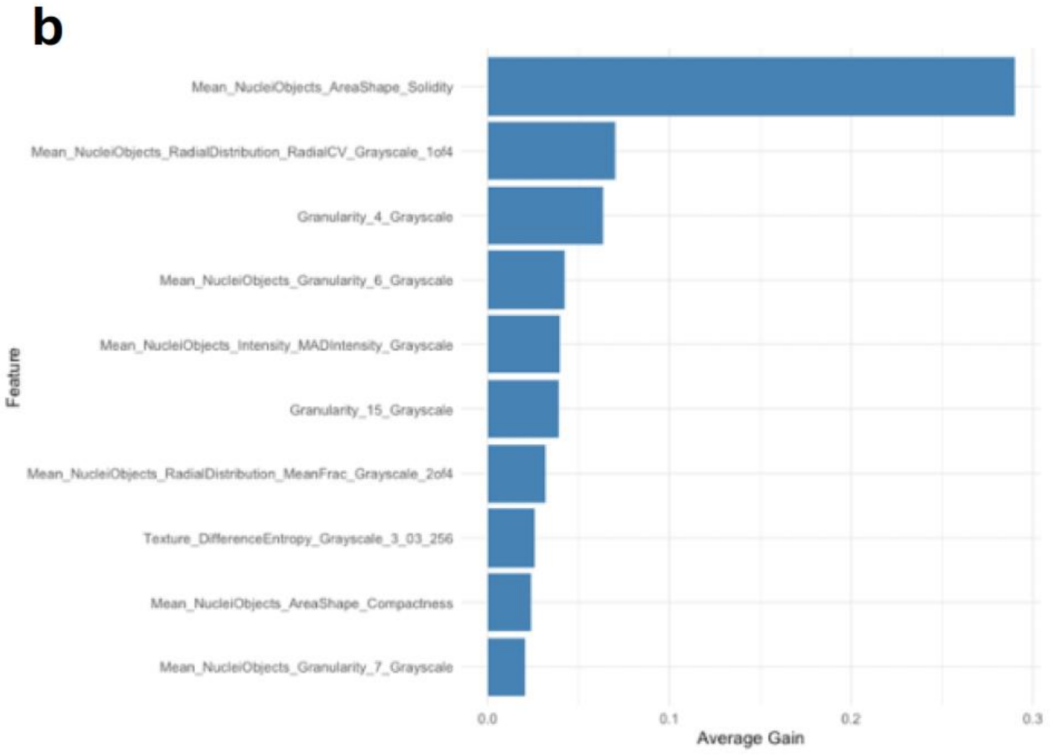

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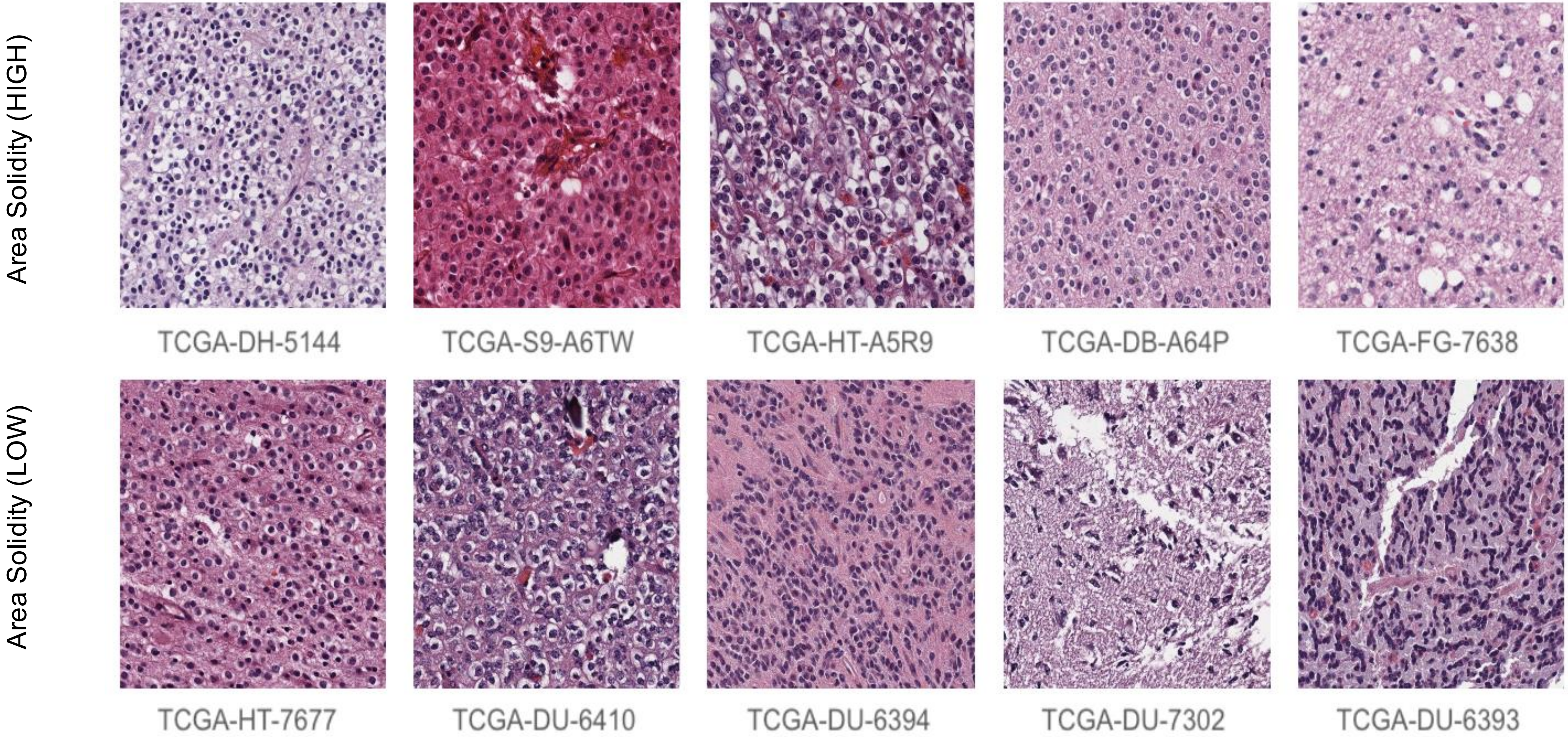
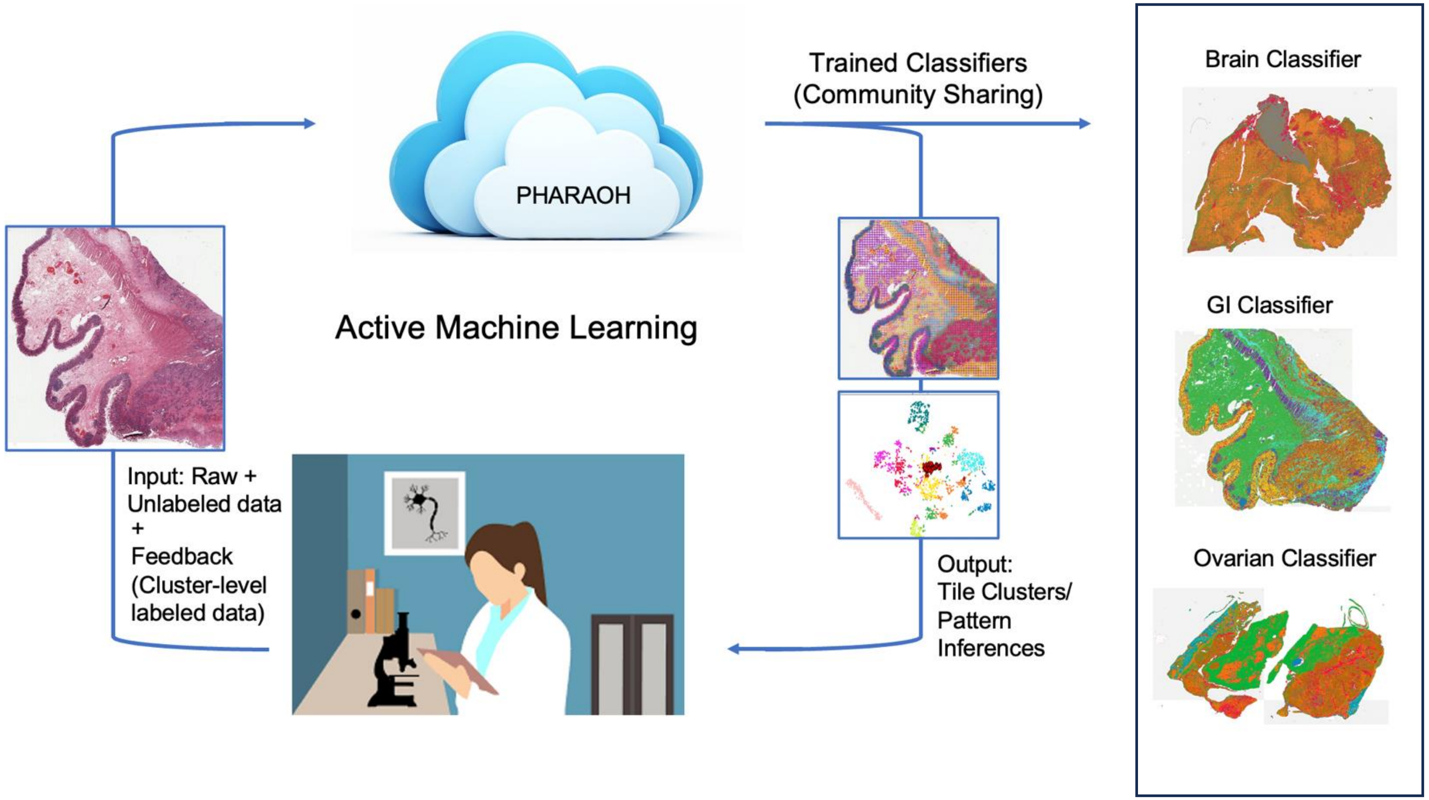


Crowdsourcing Classifier Production

Biomarker (Feature) extraction



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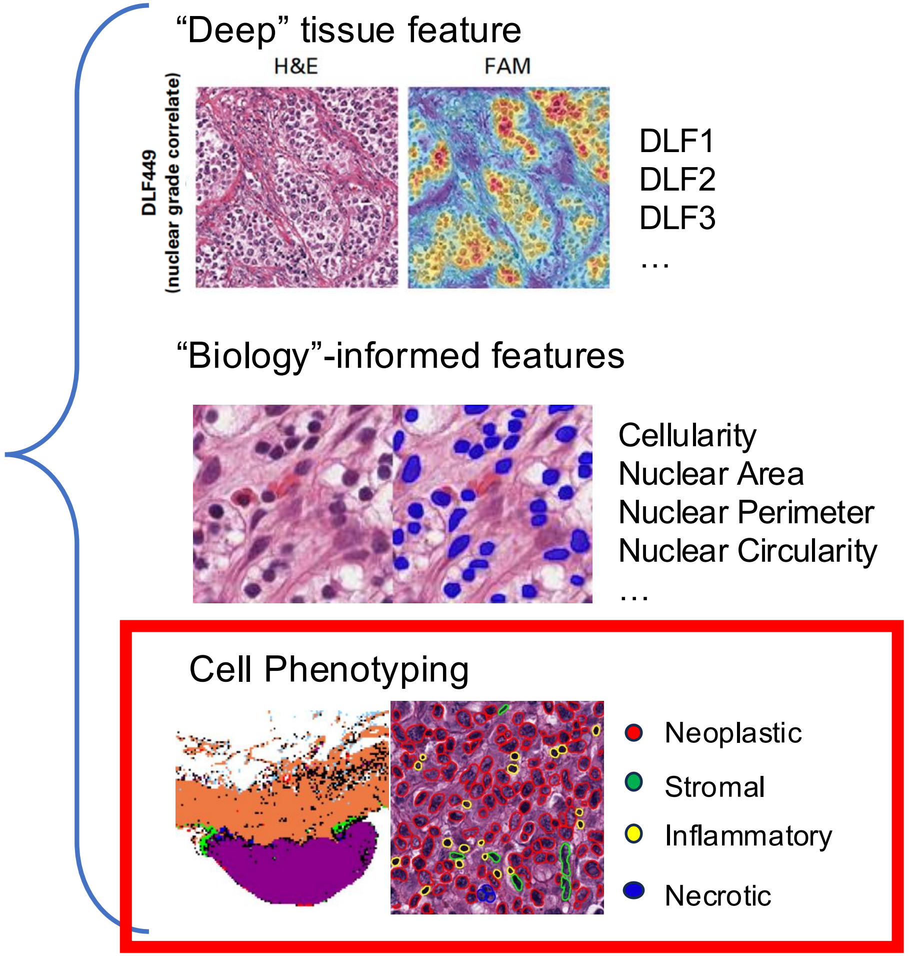
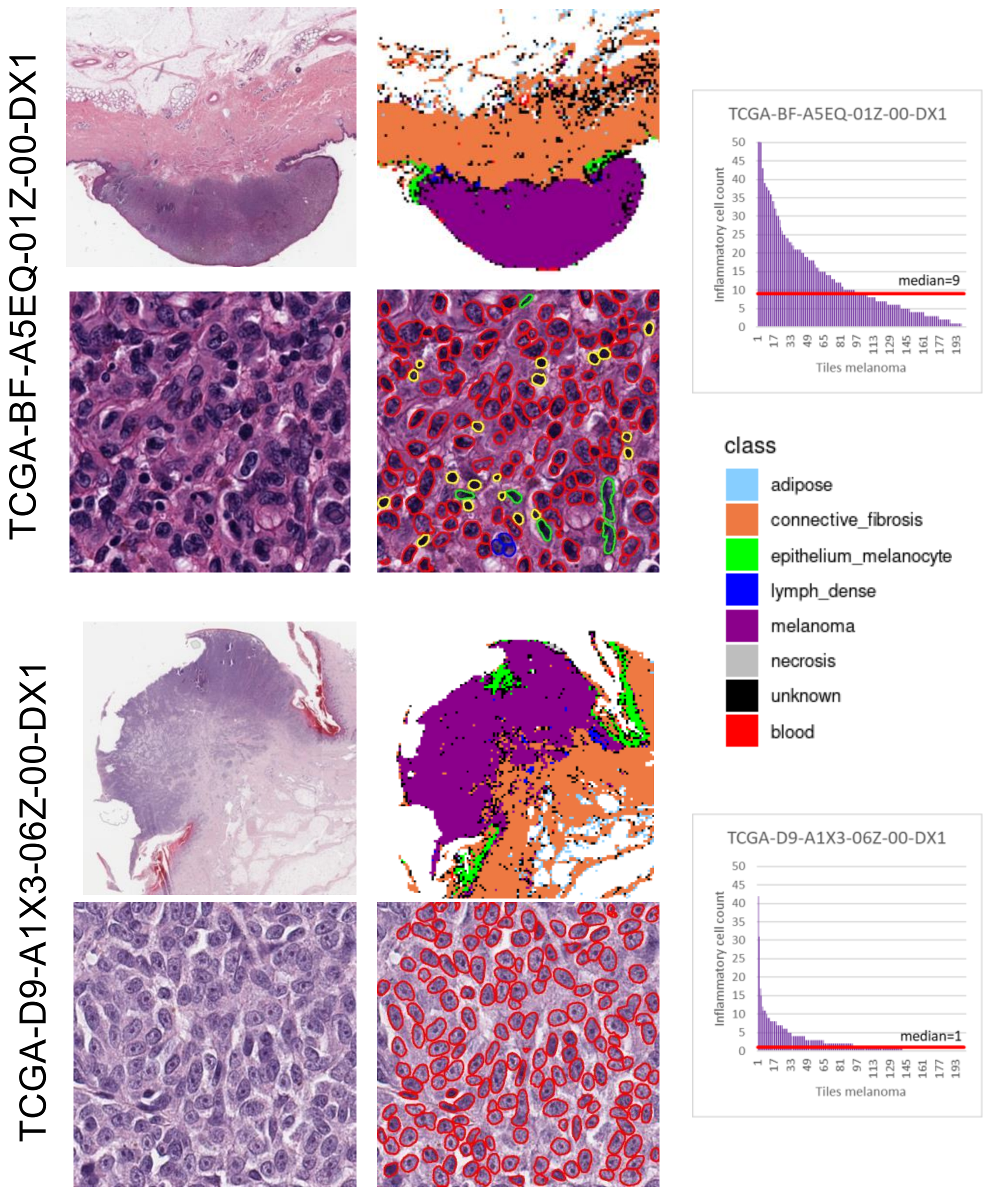
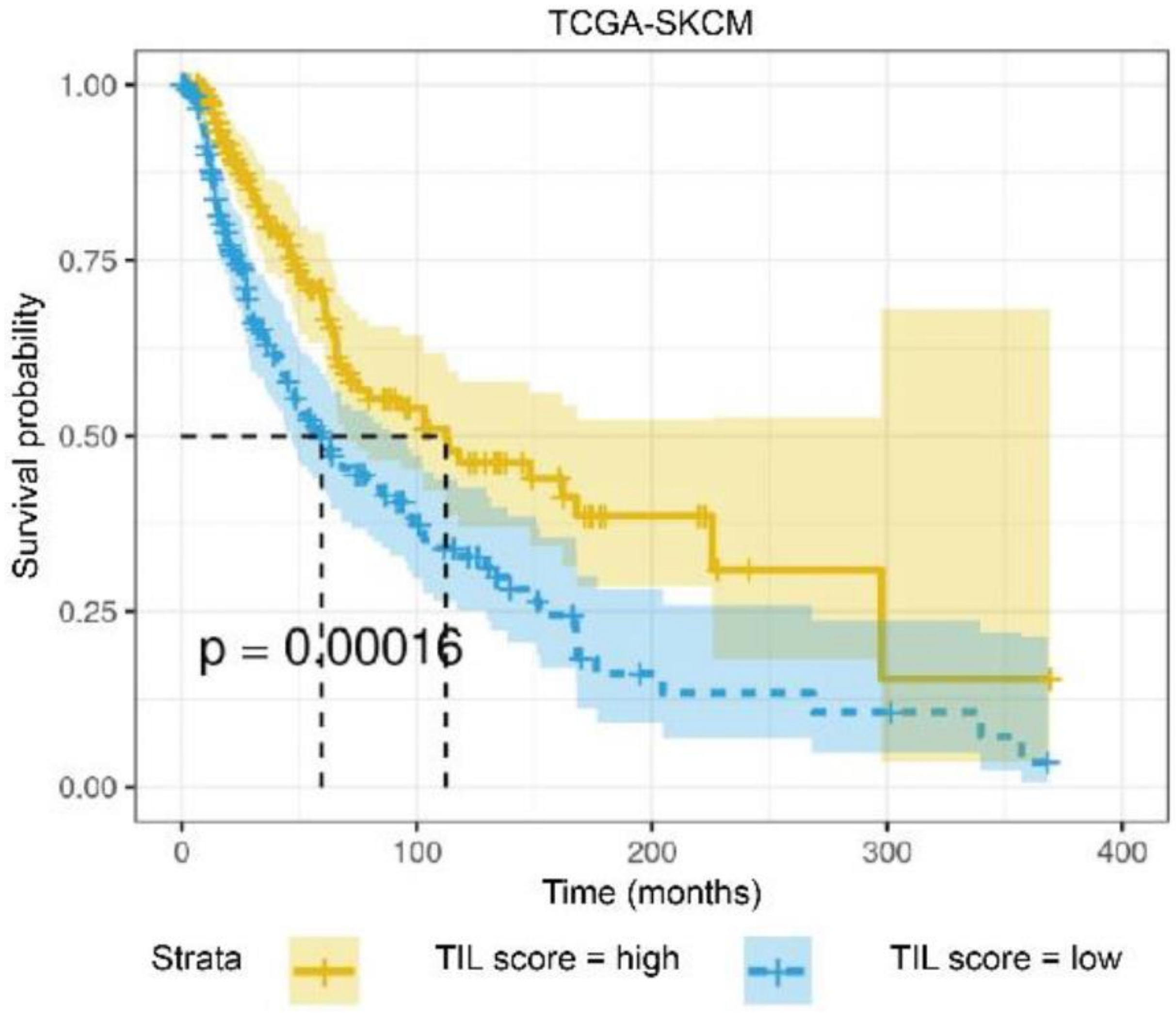


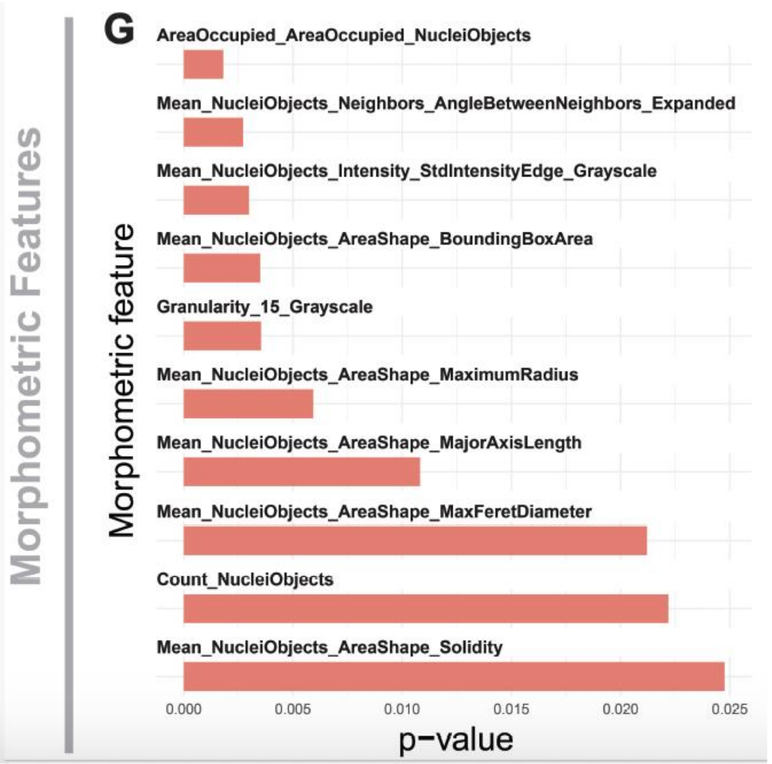
PHARAOH: Quantitative prognostic biomarker design



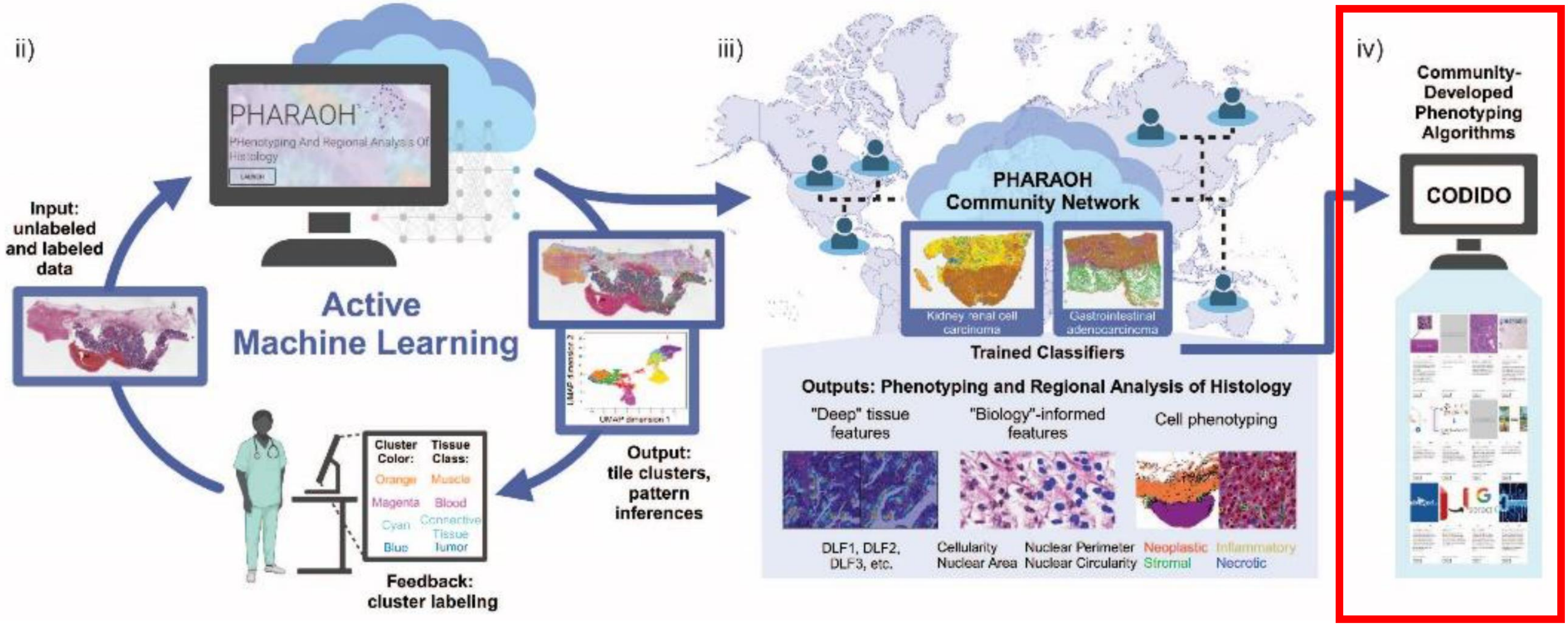


PHARAOH: Biological exploration and discovery

Mitotic Spindle Activity

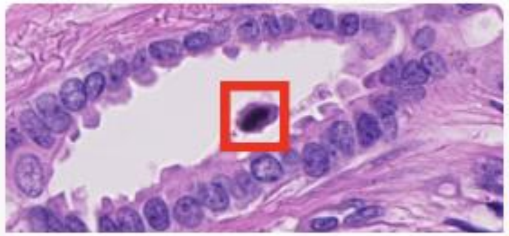


PHARAOH: Crowdsourcing Clinical Expertise



Faust & Chen et al, *Nature Communications*, 2025

Scaling Computational Pathology through crowdsourcing (codido.co)
Mitosis Detection



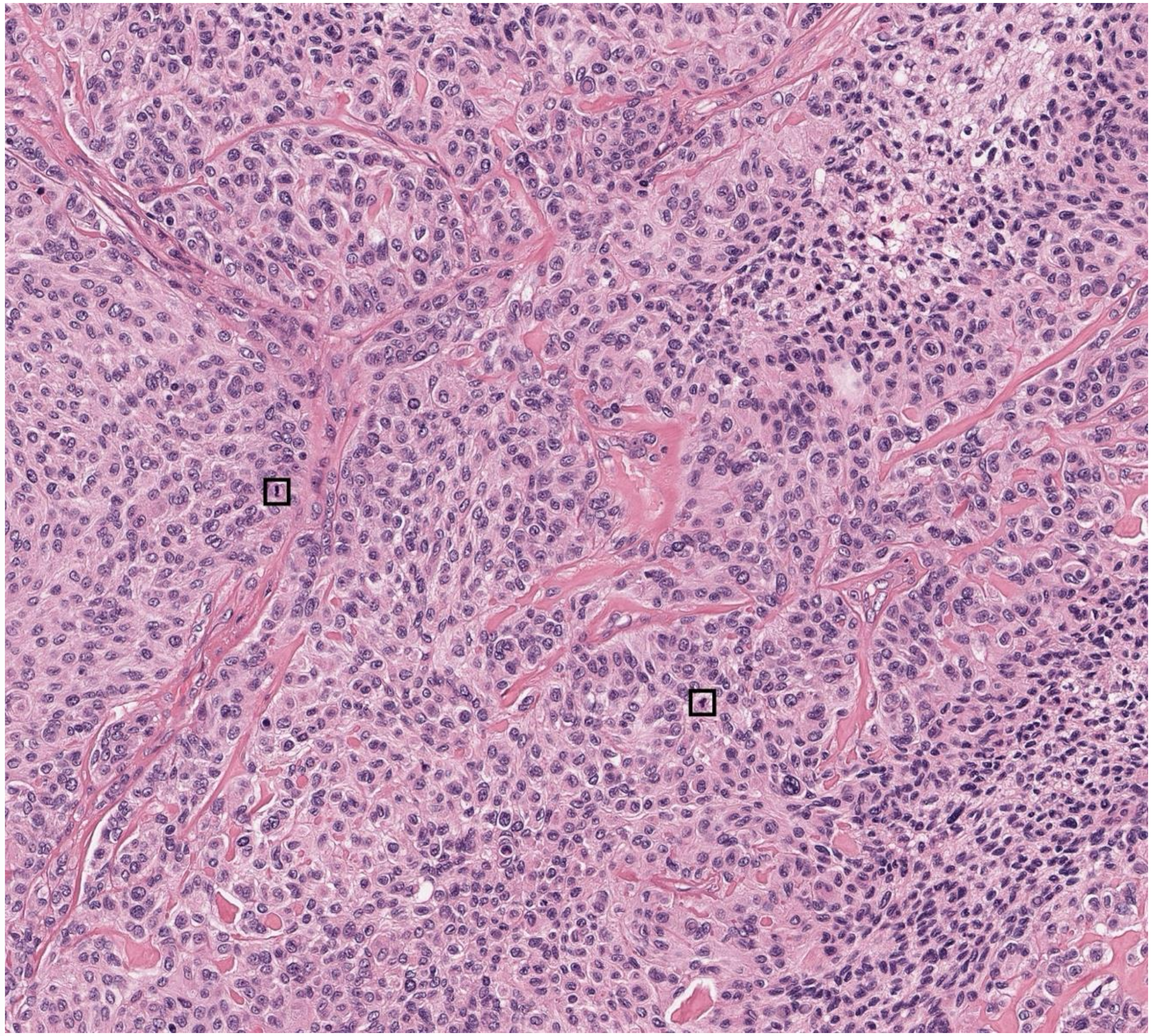
0 60

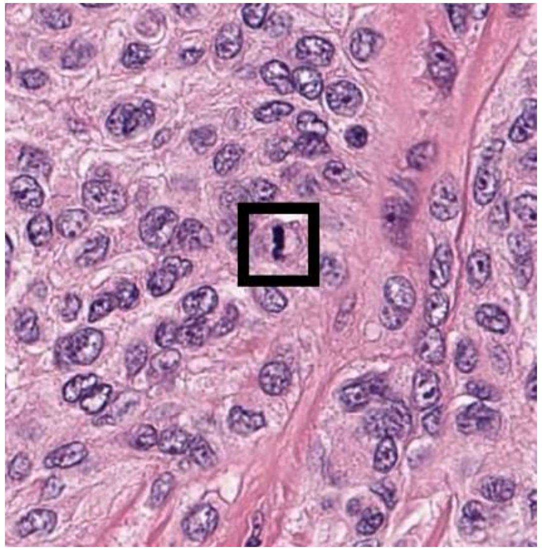
mitotic figure count

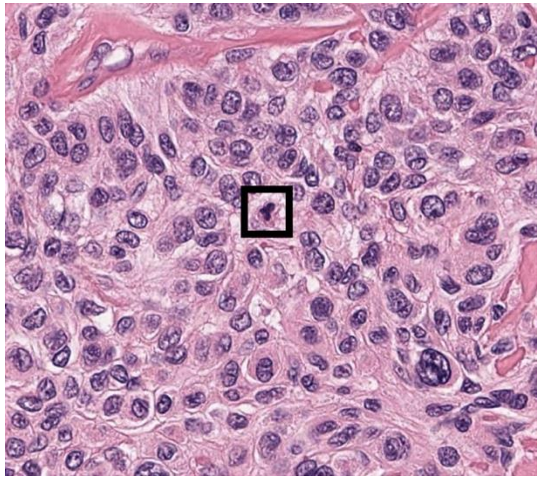
input image or zip with multiple image files. mpp = microns per pixel mm2pfov = mm2 per feild of view based on: https://www.nature.com/articles/s41597-023-02327-4 https://github.com/DeepMicroscopy/MIDOGpp/tree/main?tab=readme-ov-file

Published by Andrew Young

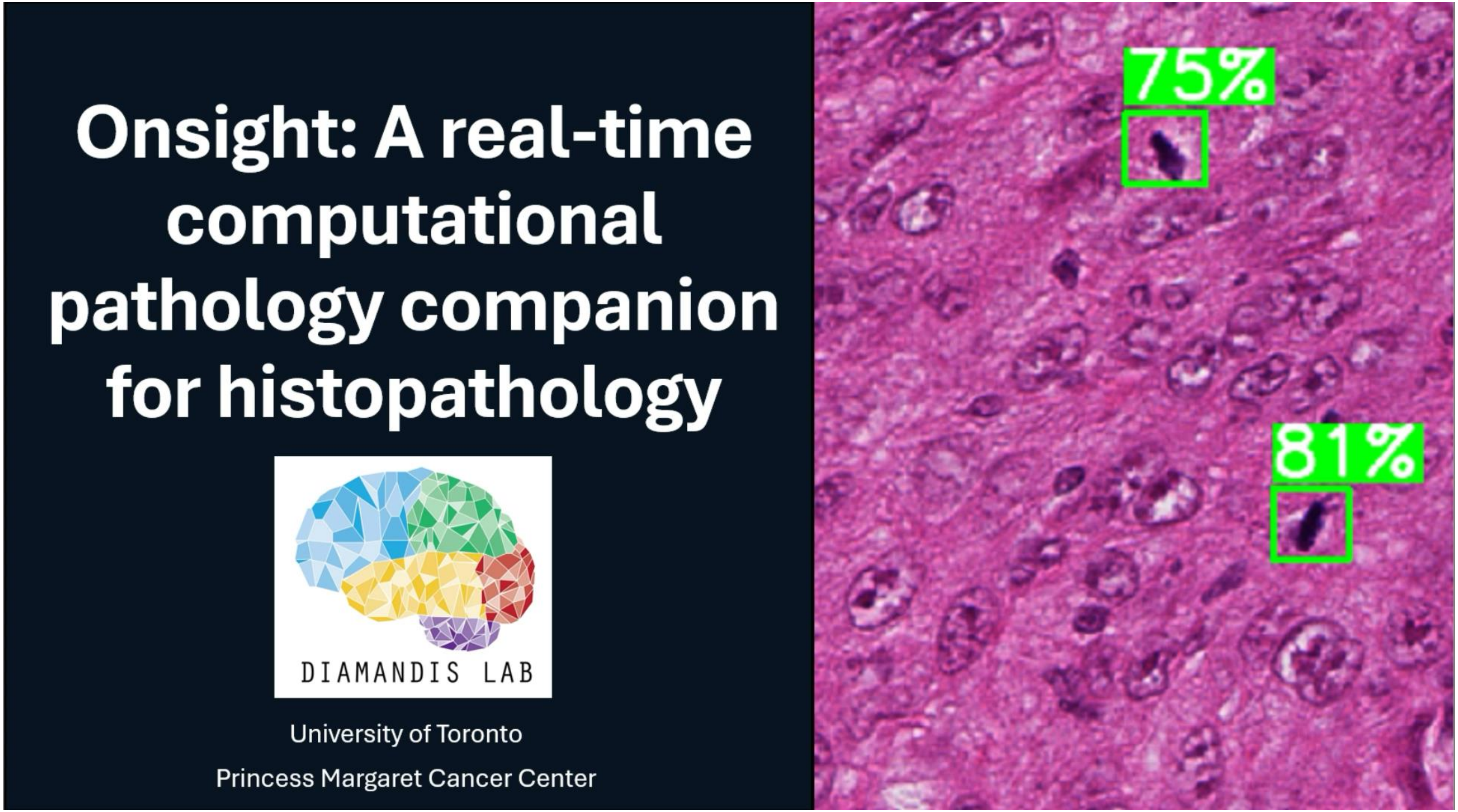
SELECT





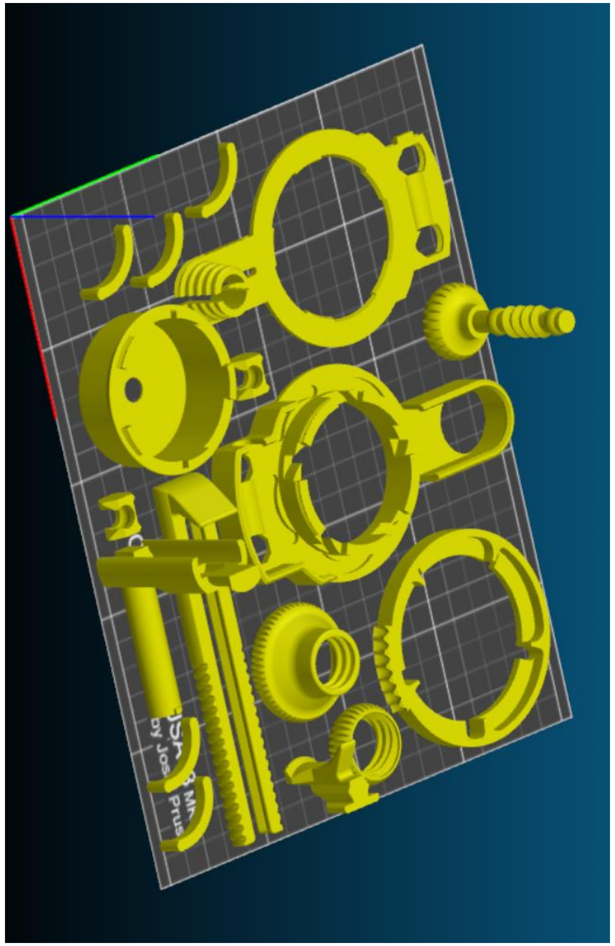


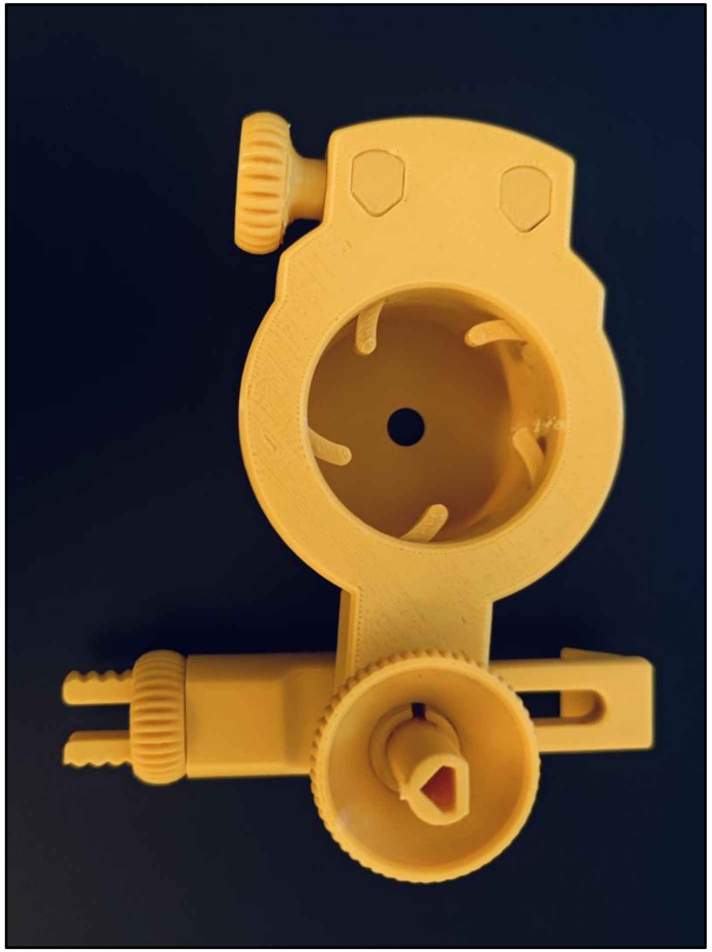
OnSight Pathology: Software for real world deployment

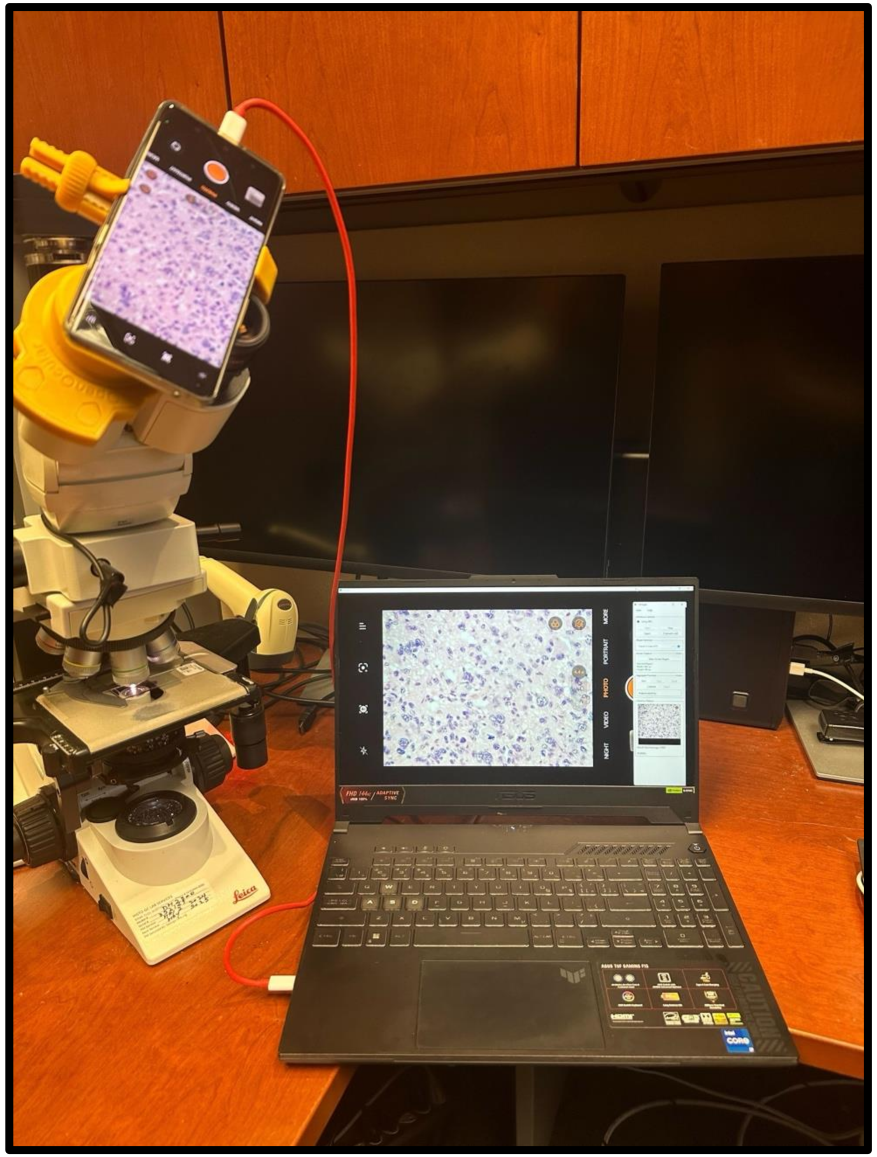


Hu et al, *Unpublished*, 2025

OnSight Pathology: Software for real world deployment

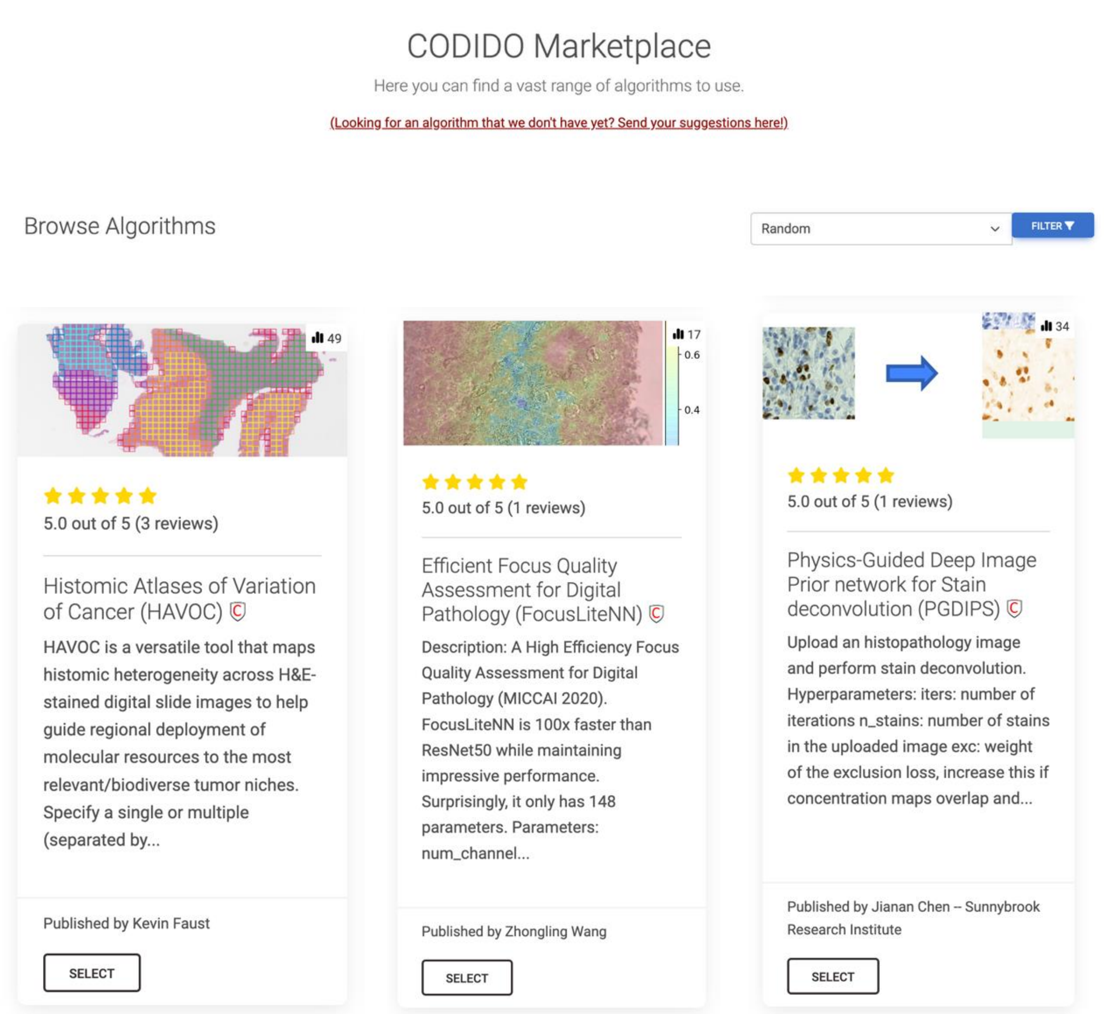


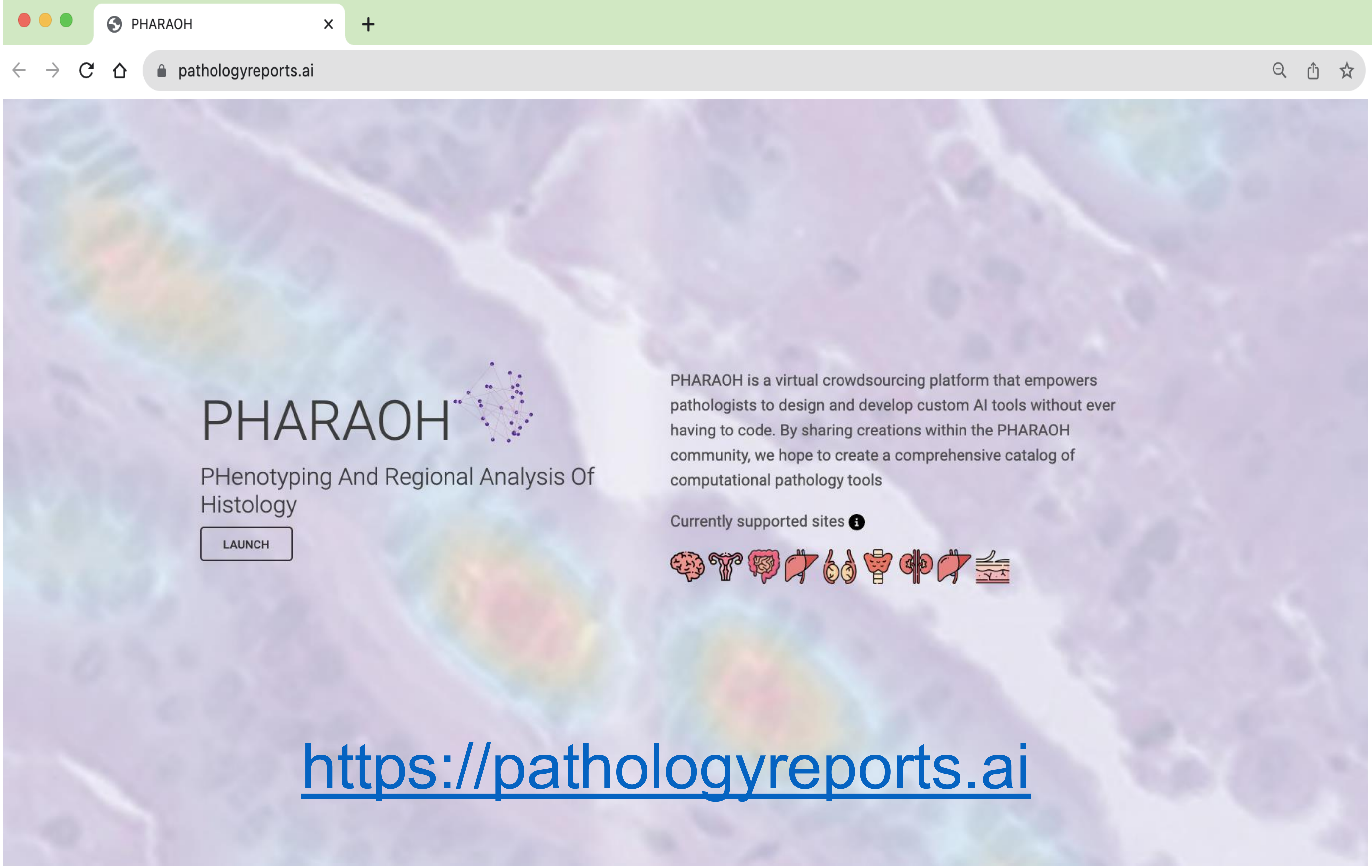




Credit: Shane Eaton PGY-4, NP

Safety Considerations / Peer-Review





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Additional Parameters / Limitations and Caveats

- Adoption has been slow compared to molecular innovations
- Caveats:
  - Staining variation
  - Magnification level
  - Context specific / Concept drift

- New Technologies/Innovations:
  - Vision Transformer (ViT) models
  - Multimodal models
  - Image Search

- Competing technologies:
  - Molecular
  - Spatial

DIAMANDIS LAB

Laboratory Medicine & Pathobiology
UNIVERSITY OF TORONTO

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Ameesha Paliwal

The BRAIN TUMOUR CHARITY

CIHR IRSC

The Princess Margaret
Cancer Foundation UHN